

Scenario

A startup is developing a conversational AI assistant for mental health support. They use pairwise comparisons from clinical psychologists to train a reward model and fine-tune the assistant using Reinforcement Learning with Human Feedback (RLHF). However, users report that the assistant often gives technically correct, but generic and emotionally flat responses that do not feel empathetic.

Q1a. What is the most likely reason for the model's failure to align with human intent?

- A. The assistant's language model was trained on outdated internet text.
- B. The reward model overfit to superficial features rather than capturing deep empathy.
- C. The training used too much GPU memory, leading to early stopping.

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[\min \left(\underline{r_t(\theta)} \underline{\hat{A}_t}, \text{clip}(\underline{r_t(\theta)}, 1 - \epsilon, 1 + \epsilon) \underline{\hat{A}_t} \right) \right]$$

`clip(·)` : Prevents **overly large updates to the policy**

“Since PPO uses the minimum between the **unclipped** and **clipped objectives**, the clipped objective should always be used. Therefore, the unclipped objective seems unnecessary.”

Explain why the unclipped objective is still necessary in PPO.

$$\underline{\hat{A}_t} = \underline{r} - \text{baseline} = 3.2 - 2.6 = 0.6$$

$$\underline{r_t(\theta)} = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} = 1.5$$

$$\underline{\epsilon} = 0.2 \quad \text{*hyper-parameter}$$

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[\min \left(\underbrace{r_t(\theta)}_{1.5} \underbrace{\hat{A}_t}_{0.6}, \underbrace{\text{clip}(r_t(\theta))}_{1.5}, \underbrace{1 - \epsilon}_{0.8}, \underbrace{1 + \epsilon}_{1.2} \underbrace{\hat{A}_t}_{0.6} \right) \right]$$

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$\text{clip}(\cdot)$: Prevents **overly large updates to the policy**

“Since PPO uses the minimum between the unclipped and clipped objectives, the clipped objective should always be used. Therefore, the unclipped objective seems unnecessary.”

Explain why the unclipped objective is still necessary in PPO. **WHAT IF $\hat{A}_t$ is negative?**

$$\hat{A}_t = \underbrace{r}_r - \text{baseline} = 2.6 - 3.2 = -0.6$$

$$\underbrace{r_t(\theta)} = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} = 1.5$$

$$\epsilon = 0.2 \quad \text{*hyper-parameter}$$

Lecture 19 & 20

Multimodal Large Language Models and Tunings

- Vision, Language, Sensors, Audio, and Beyond

** ACM MM 2024 (Rank A*) and AAAI 2025 (Rank A*) Presented Tutorial*
** Selected as the Best Tutorial in ACM MM 2024*

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Semester 1, 2025
School of Computing and Information Systems
University of Melbourne



Lecture 19 & 20: Multimodal Large Language Models and Tunings

1. Introduction and Overview

- Multimodal learning
- Importance and Challenges of integrating multiple modalities (vision, language, sensors, audio)

2. Foundation of Multimodal Large Language Models (MLLMs)

- Multimodal Key Tasks and Benchmark datasets for multimodal tasks
- Challenges in Dataset Creation and Annotation for Multimodal Integration

3. Core Techniques for Multimodal Large Language Models (MLLMs)

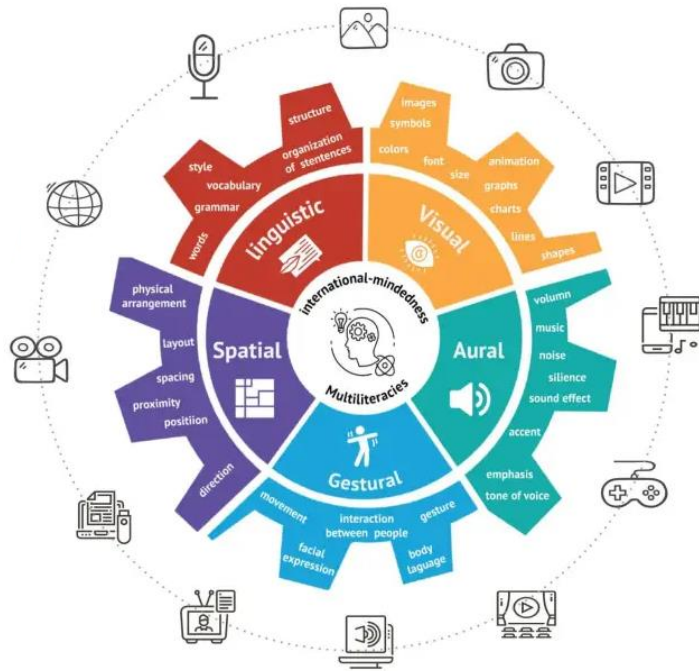
- MLLM Architecture
- MM capabilities

4. Multimodal Tunings

- Basic Instruction Tuning Technique to LVLM/MLLM Instruction Tuning
- Multimodal Chain-of-Thought
- Efficient Instruction Tuning Strategies

5. Knowledge Distillation with Multimodality

What is Multimodality?



In the context of AI and Machine Learning

*The ability of models to process, understand, and generate data from **multiple sources or "modalities,"** such as **text, images, audio, video, and sensor data**, simultaneously.*

1 Introduction and Overview

Why do we need multimodality?



Audio



Why do we need multimodality?



Audio



Why do we need multimodality?



Audio



Vision



Why do we need multimodality?

If we want AI to understand the world like we do, we must teach it to do the same. That's why **we need multimodality**.



Humans do not rely on just one sense—we combine them seamlessly.



Audio



Vision

Why do we need a multimodality?

Enhanced Contextual Understanding

Richer Information Capture:

*By combining inputs from multiple sources (e.g., vision and audio), MLLMs capture context that would be lost with a single modality, improving **comprehension and response quality**.*

More Accurate Inferences:

*Enhanced ability to detect underlying **meanings, sentiments, or actions** when **interpreting a situation with multimodal data** (e.g., distinguishing sarcasm in text combined with visual expressions).*

Personalised and Context-Sensitive Interactions:

*Particularly useful in applications where context is essential for understanding, such as **personalised assistants and healthcare diagnostics**.*

What do we discuss today?

Objectives of this lecture

1. Understand the **fundamentals of Multimodal Large Language Models (MLLMs)**



2. Explore the **challenges and potential solutions** to deal with multiple modalities

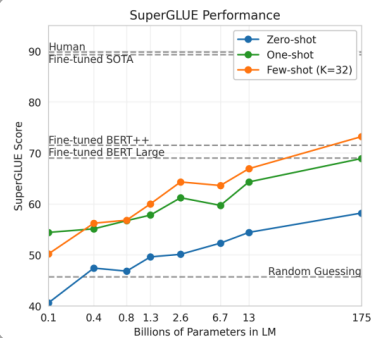


3. Discover **distillation the knowledge from different MLLMs**



What we have learned with Large Language Model!

Prompt Engineering (In-context Learning)



Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

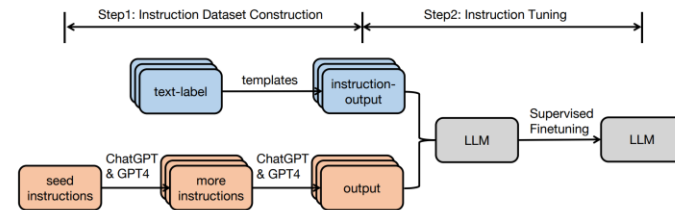
1 Translate English to French: task description
2 cheese => prompt

Few-shot

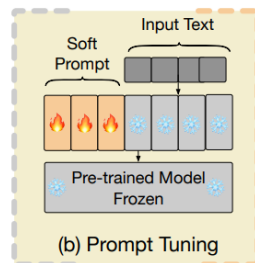
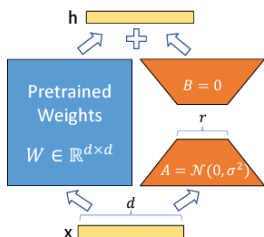
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

Translate English to French: task description
sea otter => loutre de mer examples
peppermint => menthe poivrée examples
plush girafe => girafe peluche examples
cheese => prompt

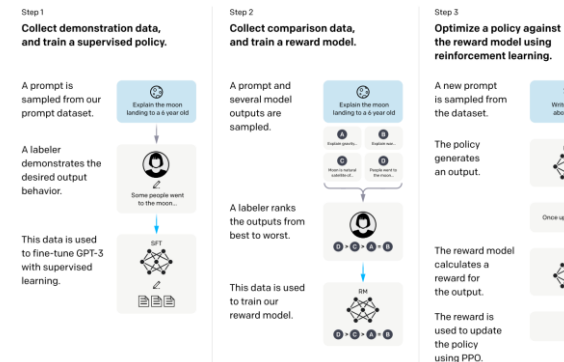
Instruction Fine-Tuning



Parameter-Efficient Fine-Tuning

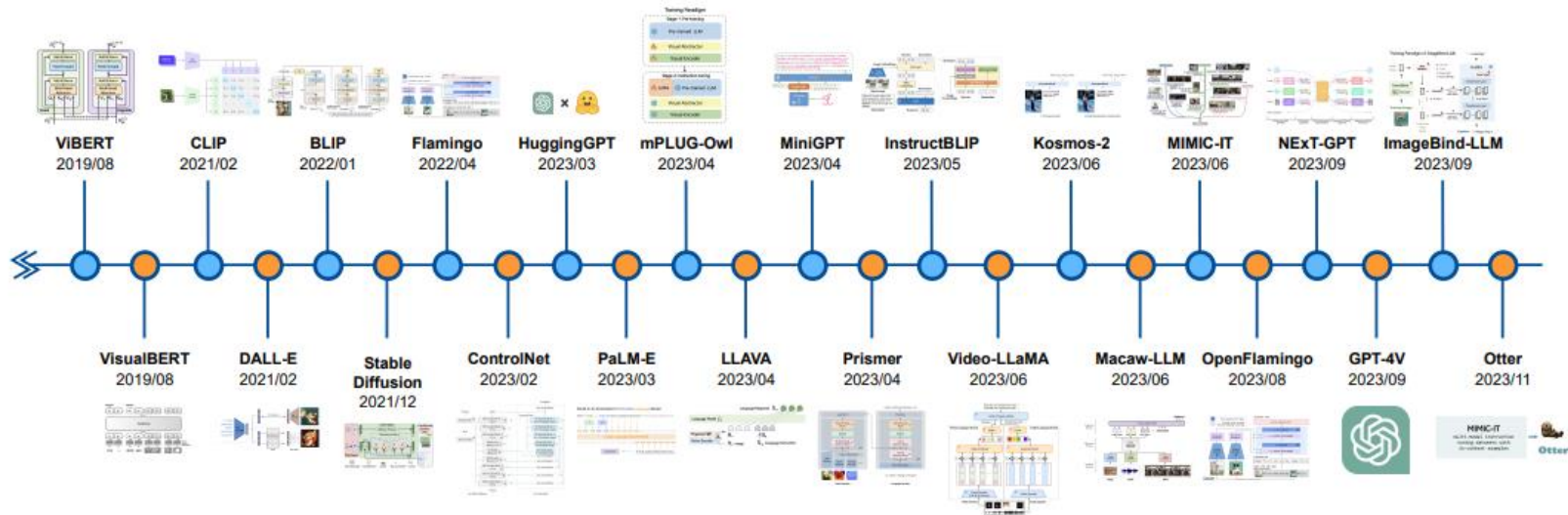


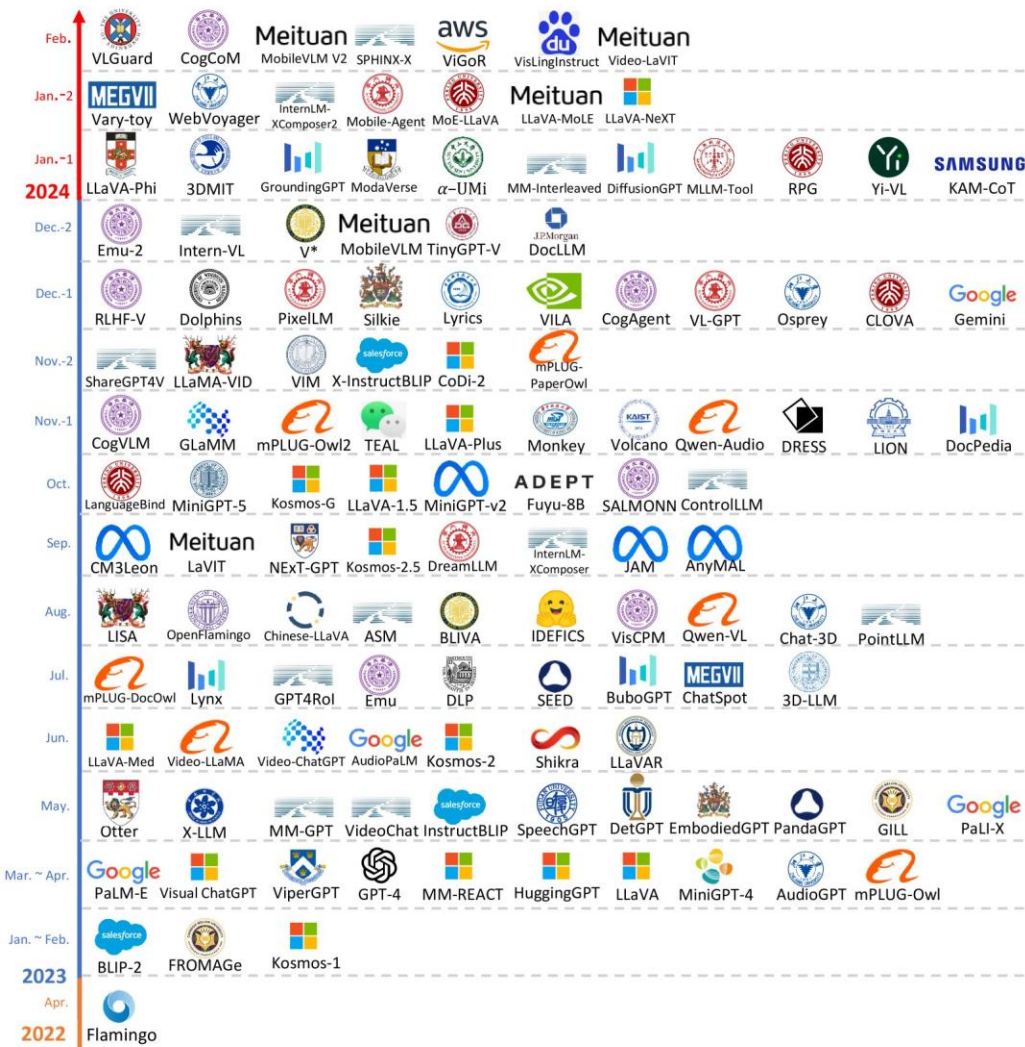
Training/Tuning with Human Feedback



Timeline of MLLMs

The exploding trend of MLLMs





Timeline of MLLMs

The exploding trend of MLLMs



Lecture 19 & 20: Multimodal Large Language Models and Tunings

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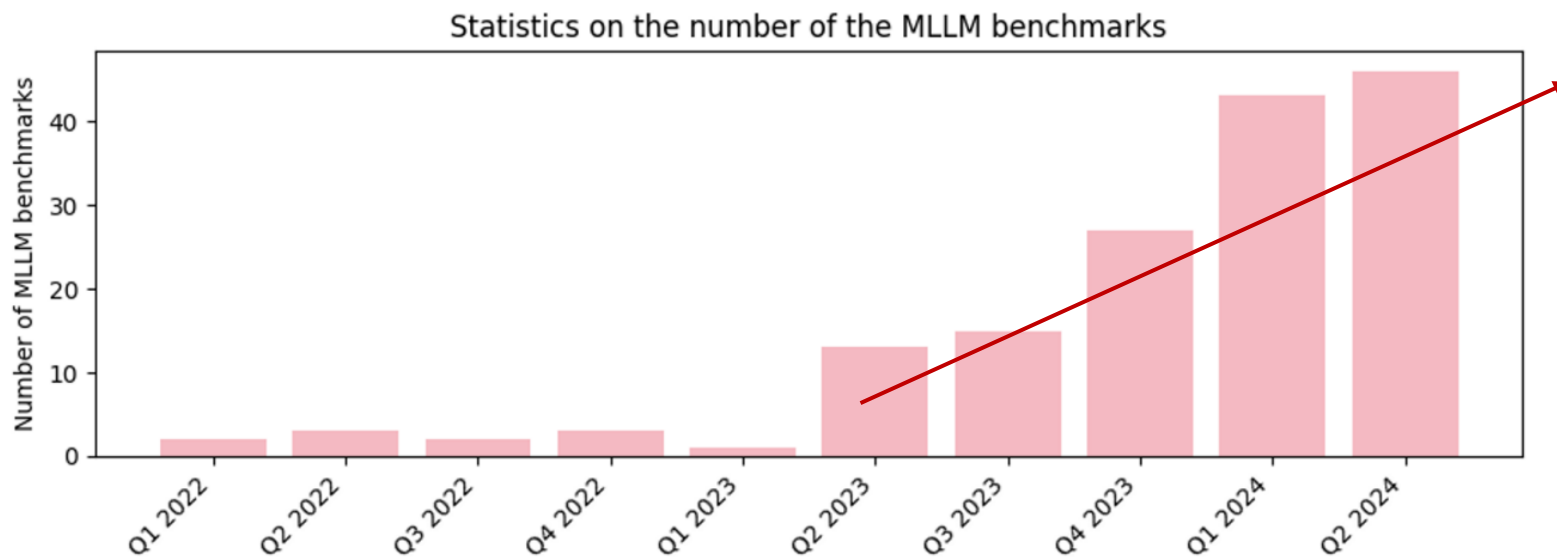
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Tasks: MLLMs Benchmarks

Trend of MLLMs evaluation benchmark papers over time



Key Tasks

The main focus of this lecture is not going through a 84 (+more) benchmarks for MLLMs.

Benchmark	Focus	Answer Annotation	Answer Type	Ref. evaluation	Model
MMBench	perception and cognition	1.5k	YN	N/A	10
MMBench-12	objective evaluation	1217	open	OPT	14
MovieChat	video understanding	14	open	OPT	6
MM-Vet	integrated capabilities	205	open	OPT	9
SciQA	scientific question answering	457k	open	OPT	5
DEMON	demonstrative instruction understanding	477.72k	open	human	9
M-HalBench	hallucination detection	160k	open	human	14
ViFi-Bench	instruction following	502	open	Human/OPT	14
EgoSchema	long video understanding	5063	MOA	N/A	4
funcBench	specific real-world functions	408	open	OPT	7
Q-Bench	low-level visual perception	2090	YN/open	N/A	15
MMHAL-BENCH	reasoning hallucinations	96	open	OPT	6
PCV-EVAL	decision-making ability	300	MOA	OPT	6
MathVista	mathematical tasks	715	MOA/open	OPT	12
MMFait	knowledge editing	N/A	YN	N/A	2
HallusionBench-M	image-context reasoning	1129	open	OPT	15
What-Vis	spatial relations	820	test options	N/A	8
CLEF	Comprehensive Evaluation Framework	—	open	OPT	9
Bingo	hallucinations related to bias and interference	170	open	OPT	2
MagnitudeBench	visual perception of small objects	249	MOA, open	Human/OPT	9
VLMAP	visual-linguistic capabilities	4177	caption matching	N/A	12
AMBER	hallucinations related to attributes and relations	136	YN, open	N/A	9
MMBench-100k	visual chains understanding	35	YN, MOA	OPT	9
Audio-Video	open ended video question answering	127	open	OPT	11
VideoBench	comprehensive evaluation for videos	1705k	MOA	N/A	4
MMU	college-level subject knowledge	11.5k	MOA/open	N/A	23
ViFiBench	video understanding	4000	MOA	N/A	17
ViFiBench	temporal concept understanding	14	open	OPT	9
MM-SafetyBench	safety-critical evaluations	5040	open	OPT	12
TimeT	long video understanding	125k	open	OPT	4
Grounding-Bench	grounding and chat capabilities	10000	open	OPT	9
EgoPlan-Bench	human-level planning	3k	MOA	N/A	28
MMBench	10 instruction following	327k	open	OPT	7
V-Bench	grounding detailed visual information	191	open	OPT	7
Event-M-DATA	continuous driving	—	MOA	N/A	21
ChartBench	chart comprehension	600k	open	N/A	21
TransportationBench	transportation	3000	MOA, YF/open	OPT	16
MM-VQA	visual perception	1910	open	OPT	10
BenchMM	robustness	—	MOA	N/A	14
MM-CAP	self-awareness in perception for MLLMs	1150	open	OPT	15
AmbiBench	image aesthetics perception	8000	MOA	OPT	15
SPEC	fine-grained comprehension	4761	test options	N/A	5
MeVisNet	segmental image reasoning abilities	4761	free-form	OPT	9
CSMM	college-level subject knowledge	14K	open	N/A	11
RTCM	real training	2000	open	OPT	10
CSMM	knowledge comprehension and reasoning	3003	open	N/A	10
Mobile-Eval	mobile device operations	23	N/A	N/A	9
LiRS-Bench	remote sensing	600	MOA	N/A	9
MMBench-L	multimodal hallucination detection	430	open	N/A	2
Celestial	celestial objects across different types	7368	open	N/A	9
SHIELD	face spoofing and forgery detection	—	TR, MOA	N/A	2
SciQA	science subjects	1045	open	OPT	6
Q-Bench	low-level visual perception	949	YN/open	N/A	28
VOQA-20K	self-awareness hallucination	20k	open	N/A	3
Acquaint	media	400	open	OPT	6
ChartQ	chart reasoning	40k	open	N/A	11
MCUB	understand inputs from diverse modalities	—	MOA	N/A	5
MMBench-15k	multi-decoding information in the prompt	1000	open	OPT	9
CORIS	context-dependent visual comprehension	706	Close&Open-ended	Human/OPT	14
MM-Soc	visual commonsense	—	MOA, open	N/A	10
ViFiBench	visual hallucination	1200	YN	N/A	8
Math-Vista	mathematical reasoning	—	open	OPT	9
MMU	knowledge editing	1000, options	open	OPT	2
MMU	inter-modality semantic coherence	—	N/A	N/A	7
OSE	other scene understanding	14084	open	OPT	8
CRPE	relation comprehension	—	MOA	N/A	3
TempComp	temporal perception	7540	multiple types of tasks	OPT	11
Hem	audio culture	1132	open	OPT	9
NHAR-VQA	reasoning abilities	900	restrictive task	N/A	9
VEKER	knowledge editing	—	open	OPT	9
CGNT	continuous instruction tuning	3174	open	N/A	2
VL-IC	in-context learning	1520	open	N/A	12
MathVista	visual math tasks	14	open	OPT	9
VisualCOT	specific local scene identification	—	free-form	OPT	3
PERB	visual scene capabilities	21.30	open	OPT	4
MMVP-Bench	visual prompting research	—	YN	OPT	4
MMU	visual independent tasks	1500	open	N/A	16
MMU	10 medical tasks	—	MOA/open	OPT	2
JaIRBench	for robustness of MLLMs against jailbreak attacks	26000	open	LLM	10
PERB	local scene behavior analysis	—	open	OPT	4
FenceUI	understanding of mobile UI screens	69	N/A	N/A	4
VisualWebBench	understanding and grounding in web scenarios	1.5k	open	N/A	14
VisualQ	real-world engineering tasks	1451	open	N/A	8
Lavi	vietnamese visual language understanding	—	open	N/A	13
UNICAM	image artistic attributes	5704	multiple types of tasks	N/A	7
CPMM	counterfactual reasoning	—	N/A	N/A	7

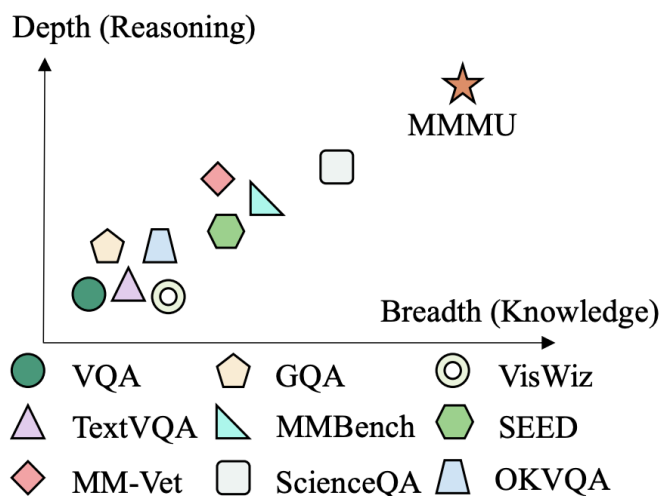
Evaluation of MLLMs

Comprehensive Evaluation	MDVP-Bench[6]; CMF[7]; UtiBench[8]; MME[9]; MM-Vet[10]; TractBench[11]; MMBench[12]; OwlEval[3]; Open-VQA[14]; SEED-Bench[5]; SEED-Bench-2[15]; LLaVA-Bench[16]; LAMB[17]
Fine-grained Perception	1. Visual Grounding and Object Detection: CODE[18]; Flickr30k; Entities[19]; Visual7W[20]; V-Bench[21]; Grounding-Bench[22] 2. Fine-grained Identification and Recognition: GVT-Bench[23]; V* Bench[21]; MMVP[24]; CV-Bench[25]; P2GR[26]; Visual-COT[27]; MagisterBench[28]; IR-Bench[29]; SPARK[30] 3. Unpaired Visual-language Alignment: Espec[31]; SPEC[32]; VALSE[33]; VL-Checkin[34]; Winogrand[35]; ARO[36]
Image Understanding	1. Multi-image Understanding: Moments[17]; MMBench[38] 2. Image Captioning: CompBench[40]; MMB[41] 3. Image Understanding: II-Bench[42]; ImplicitAI[43]; PAIR-Bench[44] 4. Image Quality and Aesthetics Perception: AestBench[45]; UNIAA[46]; DesignProbe[47]; Q-Bench[48]; Q-Bench[49]
General Reasoning	1. Visual Reasoning: MMR[50]; What'sUp[51]; GSR-BENCH[52]; CRPE[53]; VSR[54]; SpatialGPT[55]; AMR[56] 2. Context-dependent Reasoning: CODIS[57]; CFMM[58]; VL-IC-Bench[59] 3. CoT Reasoning: SCIENCEQA[60]; VisualCOT[27]; MC-COT[61] 4. Vision-independent Capabilities: CLEP[62]; VQA-20K[63]; QANAL[64]; MMSur[65]
Knowledge-based Reasoning	1. Knowledge-based Visual Question Answering: KB-VQA[66]; VQA[67]; OK-VQA[68]; A-OKVQA[69]; SOK-Bench[70] 2. Knowledge Editing: MME[61]; MIKE[72]; VLKER[73]; MC-MKE[74]
Intelligence&Cognition	1. Intelligent Question Answering: RAVEN[75]; MARVEL[76]; VCG-Bench[77]; MGLA[78] 2. Mathematical Question Answering: MathVista[79]; MathVista[80]; NP-hardVQA[81] 3. Multi-modal Question Answering: MMBench[82]; MMBench[83] 4. Multi-disciplinary Question Answering: MME[84]; CMML[86]; ScienceQA[60]; MMB[87]; CMML[88]; SciQA[89]; MULTIQ[90]
Text-rich VQA	1. Text-oriented Question Answering: OC-Bench[91]; P2GR[26]; TextVQA[92]; TextCap[93]; SEED-Bench-2-Pair[94] 2. Document-oriented Question Answering: STDS-VQA[95] 3. Table-oriented Question Answering: TableVQA[96]; TableVQA[97]; TableVQA: MM-NAR[99] 4. Chart-oriented Question Answering: ChartQA[100]; ChartQA[101] 5. ChartBench[102]; SciQA[103]; MMBench[104] 6. ChartVQA[105]; CHRPINLLM[106]; SciBench[107] 7. WebCode[108]; VisualWebBench[109]; PlotCode[110]
Decision-making Agents	1. Embodied Decision-making: VisualAgentBench[111]; EgoPlan-Bench[112]; PCA-EVAL[113]; OpenRLAI[14] 2. Mobile Agency: Mobile-Eval[115]; Free-UI[116]; CRAIR[117]
Diverse Cultures&Languages	1. CMMU[88]; Bino[118]; LLaVA-Bench[119]; MTVQA[120]; CVQA[121]; CMMU[88]; MULTIQ[90]
Other Applications	1. Geography and Remote Sensing: LiRS-Bench[122]; Charting-New Territories[123] 2. Medicine: CMAA-MMBench[124]; MML[125]; AestBench[126]; MultiMed[127] 3. Locality: ViFi[129]; MM-Soc[129]; TransportationGen[130] 4. Accessibility: MMB[131]; DesignAI[7] 5. Autonomous Driving: NuScenesQA[132]; DriveML-DATA[133]
Conversation Abilities	1. Long-context: Mito-Bench[38]; MMBench[134]; MMLU[135] 2. Instruction Following: GIN[36]; MIA-Bench[137]; DEMON[138]; ViFi-Bench[139]
Hallucination	POPE[140]; GAVR[141]; HaLLM[142]; M-HalBench[143]; Bingo[144]; HallusionBench[145]; ViFi[146]; CorrelationQA[147]; CHAIR[148]; MIA-Bench[149]; VideoHalluc[150]; MMBHAL-BENCH[151]; AMBER[152]; MMECaption[153]; MMBal-Bench[151]
Trustworthiness	1. Robustness: MAD-Bench[154]; MMR[155]; MM-SpBench[156] 2. Safety: MMU-Bench[160]; Jailbreak-V2[161]; MultiTrust[162]; MM-SafetyBench[163]; SHIELD[164]; RVTLM[165]
Video	1. Temporal Perception: MVBench[166]; TimeT[167]; VLMA[168] 2. Video Understanding: TempComp[170]; OCR[171]; ARMA[172]; Perception Test[173] 3. Long Video Understanding: MovieChat[174]; EgoSchema[175] 4. Video Understanding: VideoBench[176]; MMBench-Vide[179] 5. Video-MME[179]; AutoEval-Video[180]; MMWorld[181]; WorldNet[182]
Audio	AudioBench[183]; MoChosMusic[184]; AIR-Bench[185]
3D Points	Scan4M[186]; ScanReason[187]; LAMB[171]; SpatialGPT[55]; MMBench[188]
Omni-modal	MCTR[189]; AVQA[190]; MusicAVQA[191]; MMT-Bench[192];



Key Tasks

So many Vision-Language focused tasks, such as Visual Question Answering, Image-to-xx, ...



Dataset	Size	Images	Format	Source	Answer
VQA	> 1M	V	I+T	Annotated	Open
GQA	> 1M	V	I+T	Synthesized	Open
VisWiz	32K	V	I+T	Annotated	Open
TextVQA	45K	OC	I+T	Annotated	MC
OKVQA	14K	V+OC	I+T	Annotated	Open
SEED	19K	V+OC	I+T	Annotated	MC
MMBench	3K	V+OC	I+T	Repurposed	MC
MM-Vet	0.2K	V+OC	I+T	Annotated	Open
ScienceQA	6K	5 Types	I+T	Textbooks	MC
MMMU	11.5K	30 Types	Interleaved	Textbooks, Internet, Annotated	Open / MC

Key Tasks

So many Vision-Language focused tasks, such as Visual Question Answering, Image-to-xx, ...

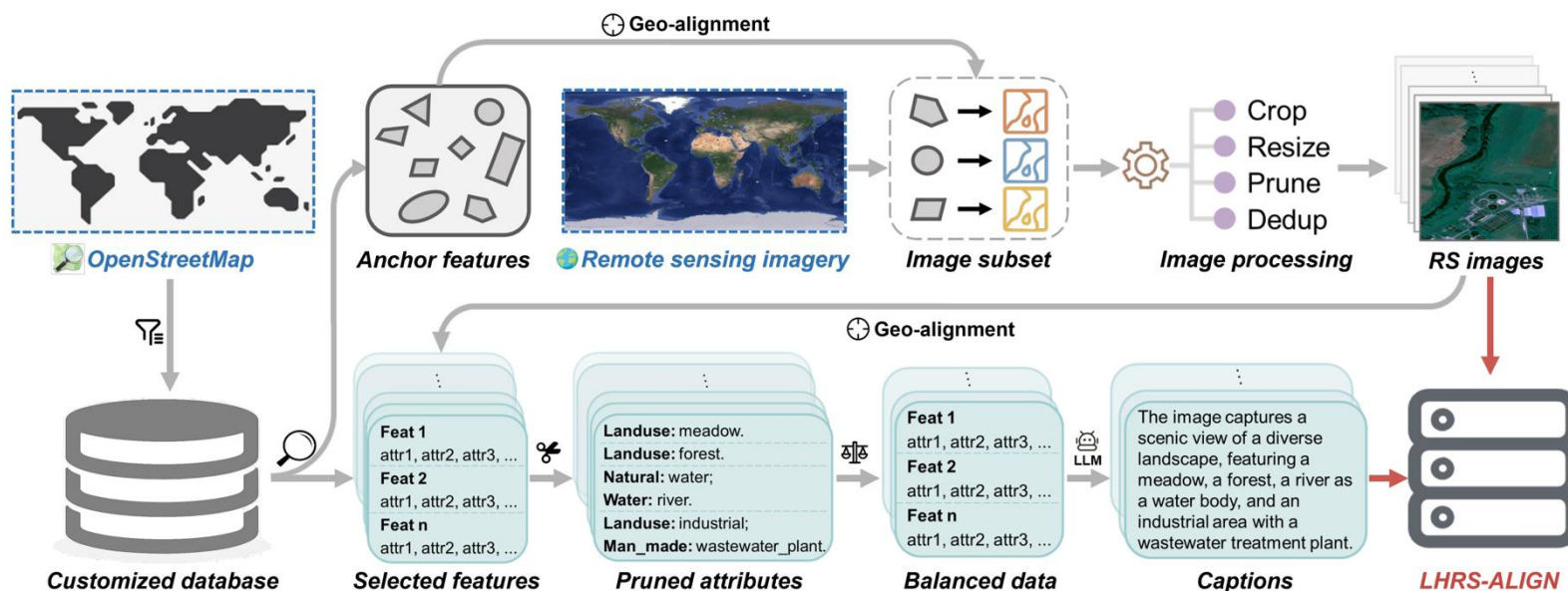
What are the recent and prominent multimodal benchmarks that leverage language?



					Answer
OKVQA	11K	V+OC	I+T	Annotated	Open
SEED	19K	V+OC	I+T	Annotated	MC
MMBench	3K	V+OC	I+T	Repurposed	MC
MM-Vet	0.2K	V+OC	I+T	Annotated	Open
ScienceQA	6K	5 Types	I+T	Textbooks	MC
MMMU	11.5K	30 Types	Interleaved	Textbooks, Internet, Annotated	Open / MC

Key Tasks: Specific Domain (Geography)

Geography – LHRS-Bot (Muhtar et al. 2024)



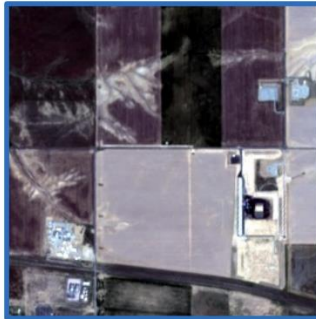
Key Tasks: Specific Domain (Geography)

Geography – ChartingNewTerritories (Roberts et al. 2024)

Satellite imagery season detection

Prompt:

These 4 satellite images are taken in different seasons. For each image, provide a prediction of the season. Select a single season per image, without replacement, from the following: Spring, Summer, Autumn, Winter. Let's think step by step.



The image shows a landscape with a lack of greenery, and the fields are mostly brown with no visible crops. This could indicate a time either before planting or after harvest. There is no snow, so it's likely not winter. This image could represent either autumn, after harvest time, or early spring, before planting has begun. **Prediction: Spring.**



The image shows fields that appear to be lush and possibly in full growth. This is indicative of a time when crops are at their peak, which is typically in the summer. **Prediction: Summer.**



The image shows fields with varying shades of brown and some patches that could be interpreted as either harvested fields or fields lying fallow. There isn't a clear indication of snow or full greenery. This image might suggest the season is either autumn, with fields that have been recently harvested, or late winter, where some vegetation is still dormant. **Prediction: Autumn.**



The image has a darker tone to it, and some of the fields have a very slight white hue which might suggest frost or light snow. This could imply that the season is winter. **Prediction: Winter.**

Test GPT-4V's ability to detect seasonal changes in a four-image time-series

Key Tasks: Specific Domain (Geography)

Geography – ChartingNewTerritories (Roberts et al. 2024)

Bounding box / Object Counting

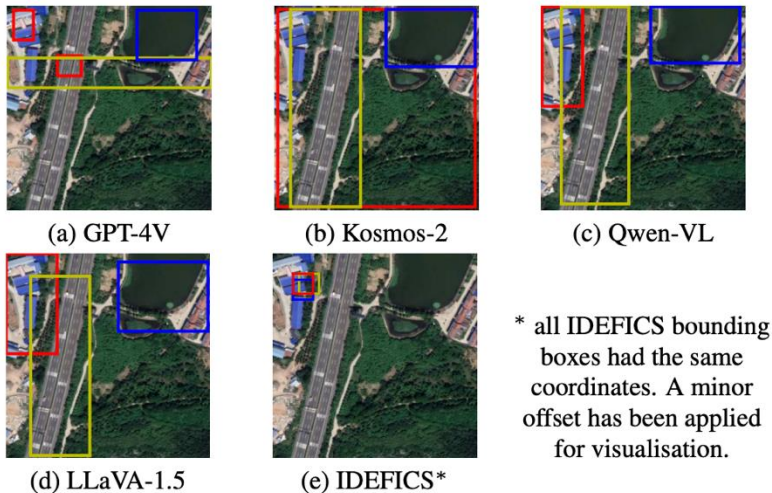


Figure 4. **Bounding boxes** for **urban areas**, **road** and **water bodies**.



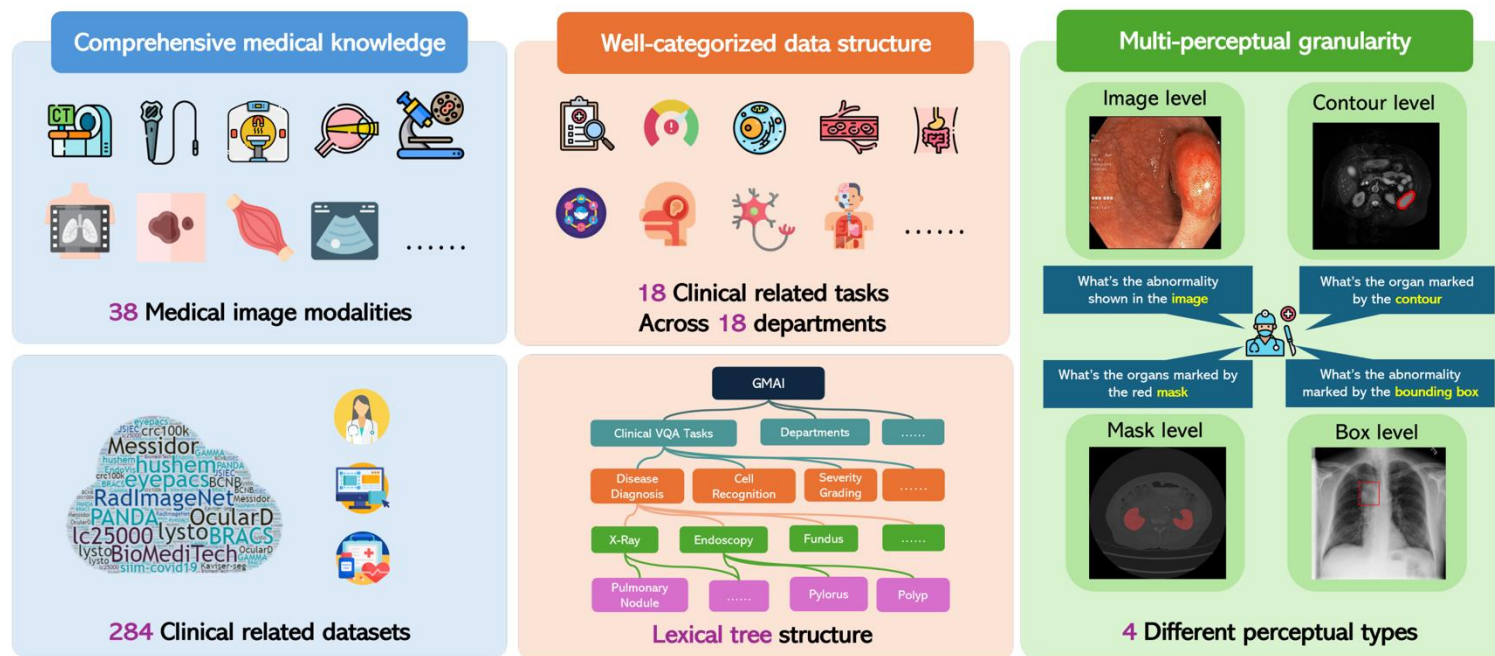
Prompt:
How many sea lions are in this image?

Answer: 45
GPT-4V: 36
IDEFICS: 10
LLaVA-1.5: 2
Qwen-VL: 0
InstructBLIP: 10

Figure 5. **Counting** small objects proves challenging.

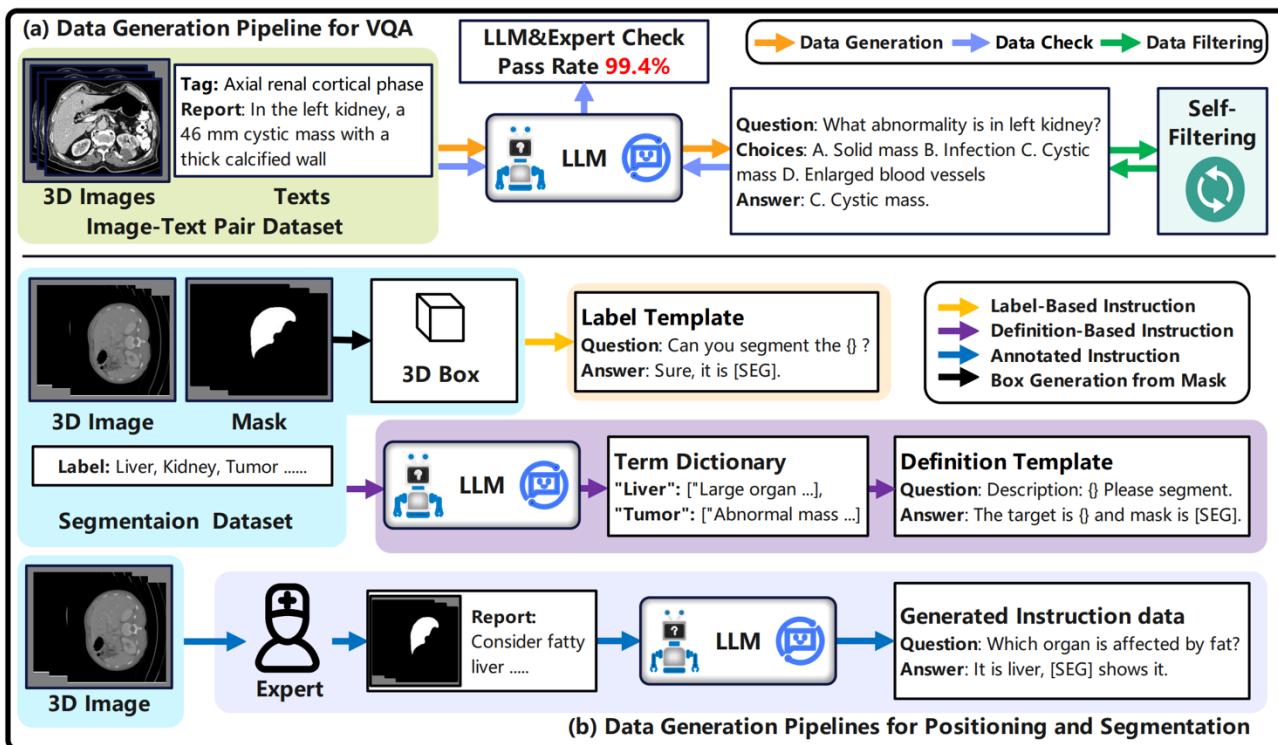
Key Tasks: Specific Domain (Medical)

Real-world clinical scenario (Chen et al. 2024)



Key Tasks: Specific Domain (Medical)

M3D – 3D Medical Image Analysis (Bai et al. 2024)



Key Tasks: Specific Domain (Medical)

M3D – 3D Medical Image Analysis (Bai et al. 2024)

Datasets	Types	Tasks	Images	Texts
VQA-Med [5]	2D	VQA	3,200	12,792
MIMIC-CXR [22]	2D	ITR, RG	377,110	227,835
PMC-OA [35]	2D	ITR, RG	-	1,646,592
PMC-VQA [73]	2D	VQA	149,075	226,946
RP3D-Caption [68]	3D	ITR, RG	51K	-
RP3D-VQA [68]	3D	VQA	-	142K
M3D-Cap	3D	ITR, RG	120,092	42,496
M3D-VQA	3D	VQA	96,170	509,755
M3D-RefSeg	3D	REC, REG, SS, RES	210	2,778
M3D-Seg	3D	REC, REG, SS, RES	5,772	149,196*

* In segmentation datasets, the number of texts can be linked to semantic masks.

VQA: Visual Question Answering under closed- and open-ended.

ITR: Image-Text Retrieval.

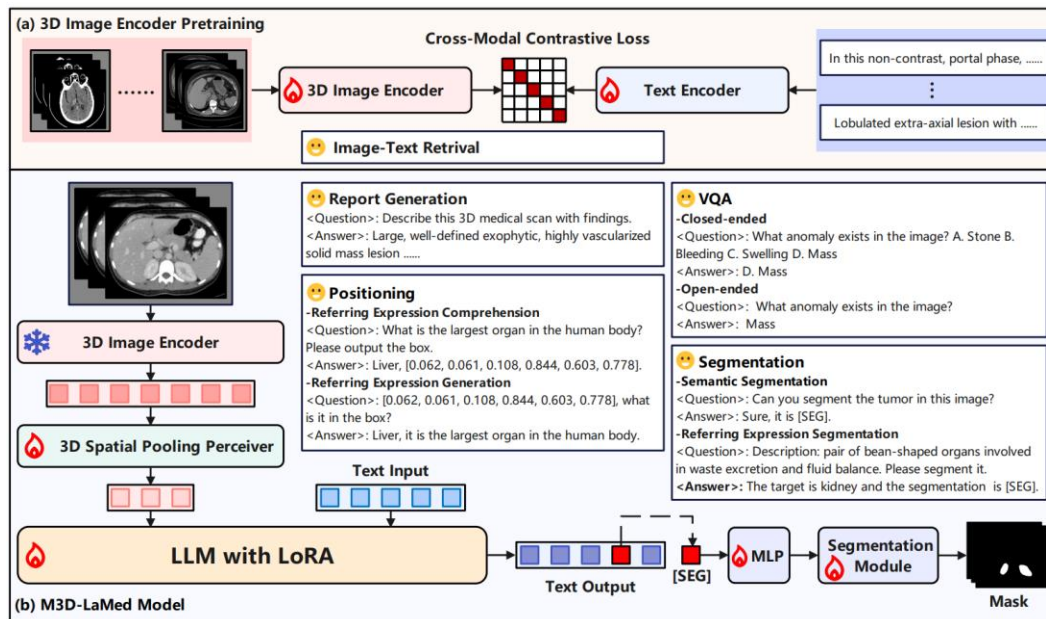
RG: Report Generation.

REC: Referring Expression Comprehension, **REG**: Referring Expression Generation,

SS: Semantic Segmentation, **RES**: Referring Expression Segmentation.

Key Tasks: Specific Domain (Medical)

M3D – 3D Medical Image Analysis (Bai et al. 2024)



VQA: Visual Question Answering under closed- and open-ended.

ITR: Image-Text Retrieval.

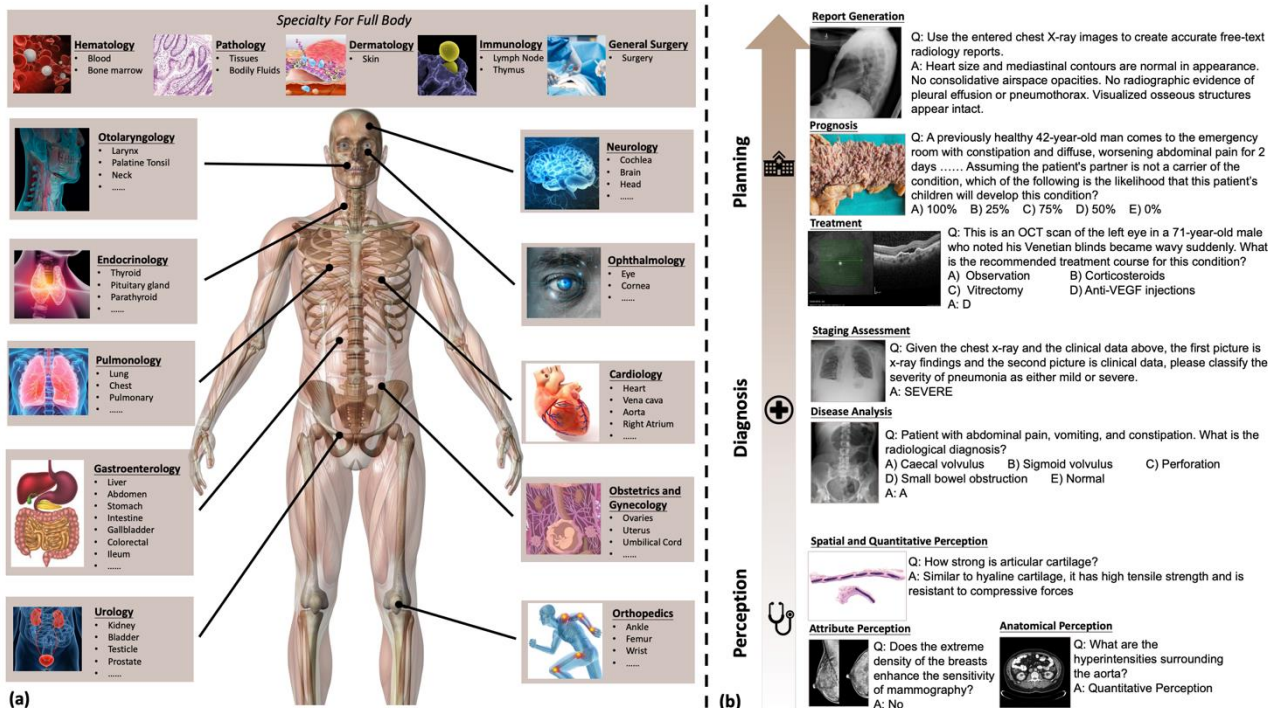
RG: Report Generation.

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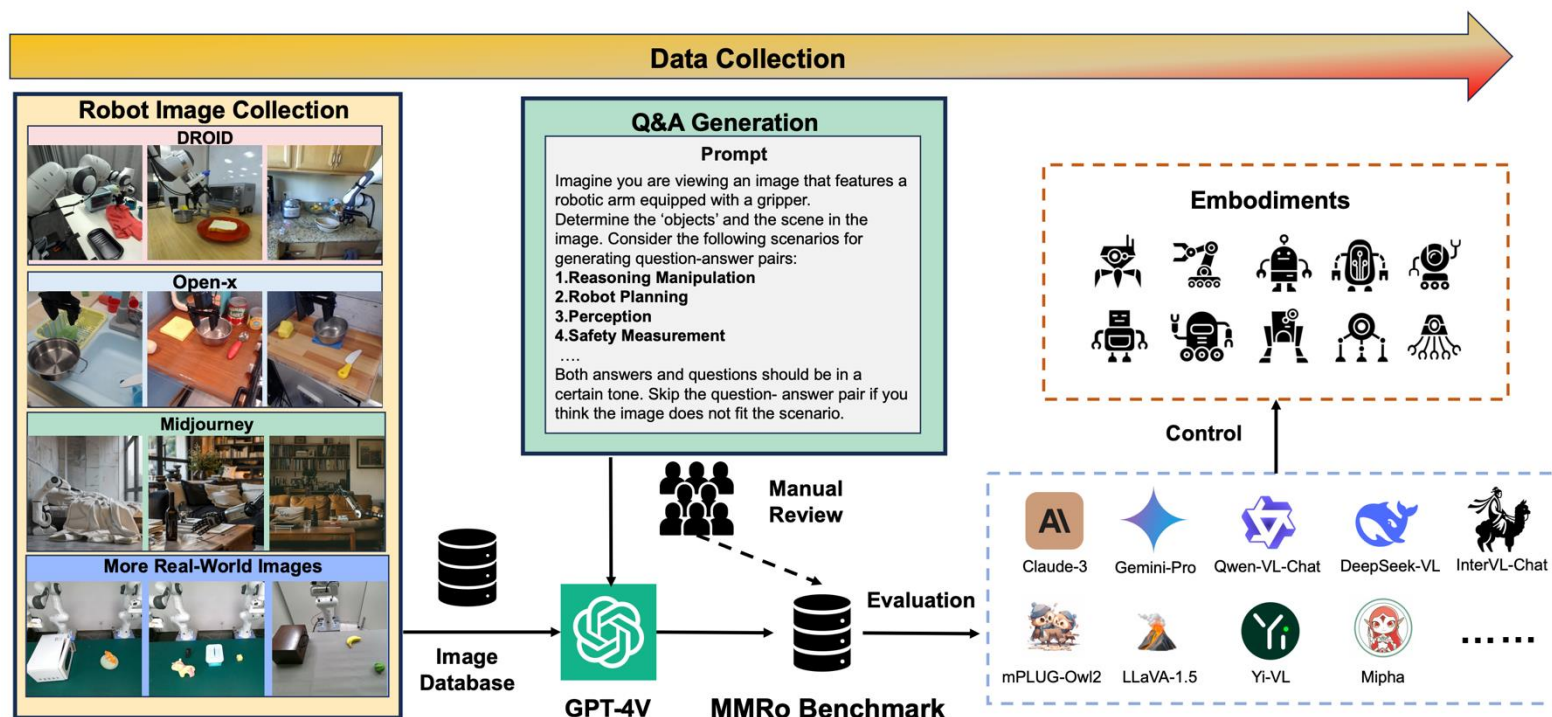
Key Tasks: Specific Domain (Medical)

Asclepius – Diverse Body parts/organ (Wang et al. 2024)



Key Tasks: Specific Domain (Robotics)

MMRo – for Robots (Li et al. 2024)



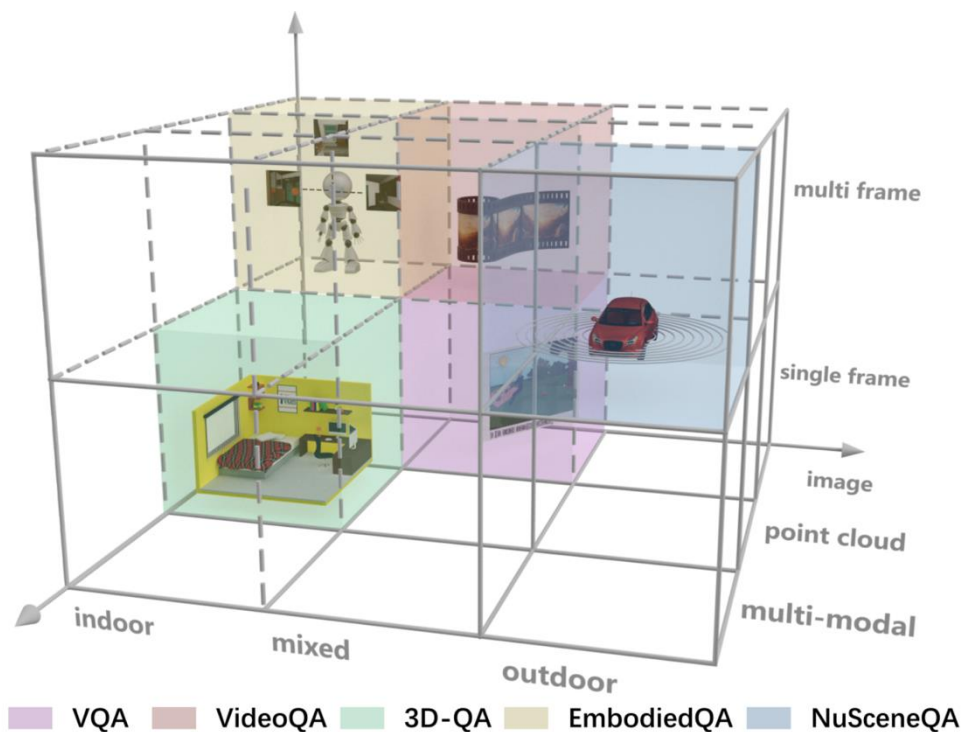
Key Tasks: Specific Domain (Robotics)

MMRo – for Robots (Li et al. 2024)

Sub-Domain	# of Q&A Pair	Task Definition
Perception		
Object Color	2174	Discriminating the color of the object.
Object Shape	2158	Discriminating the shape of the object.
Material Composition	2156	Identifying the material of the object.
Object Counting	2185	Counting the number of specified items in the scene.
Object Orientation	2116	Detecting the position and orientation of objects.
Visual Grounding	803	Return the bounding box of the queried object.
Task Planning		
Task Sequencing	2176	The ability to decompose a plan into a proper sequence of sub-tasks.
Spatial Awareness for Object Manipulation	2056	Evaluate the decision-making ability in the presence of interfering substances.
Visual Reasoning		
Object Function Identification	2178	The ability to understand the functionality of the object.
Tool Appropriateness	2173	Determine whether the queried object is a suitable tool to perform the task.
Anticipation of Action Consequences	2179	Anticipating specific outcomes of a robot action.
Safety Measurement		
Sharp Object Handling	919	Whether the object is sharp and needs careful manipulation.
Delicate Object Manipulation	2101	Determine if the object is delicate for grasping.
Heat Safety Protocol	801	Determine whether the object could have a high-temperature surface.
Total	26175	-

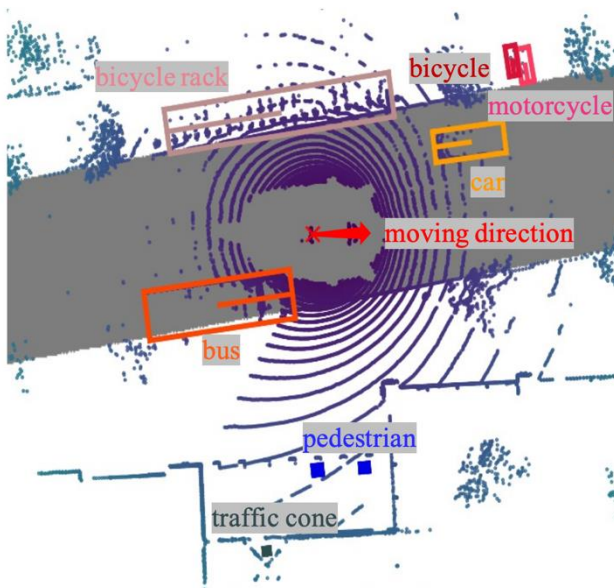
Key Tasks: Specific Domain (Autonomous Driving)

NuScenes-QA- Driving (Qian et al. 2024)

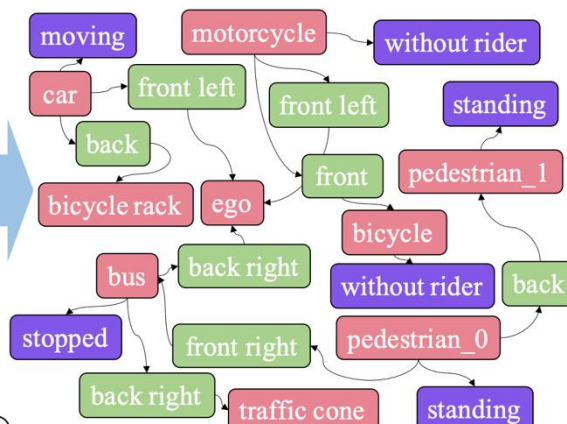


Key Tasks: Specific Domain (Autonomous Driving)

NuScenes-QA- Driving (Qian et al. 2024)



Step ①
Scene
Graph



Step ②

Question Template Construction

What is the <A2> <O2> to the <R> of the <A> <O>?
Are there any <A2> <O2>s to the <R> of the <A> <O>?

Step ③

Q&A Pair Instantiation

Q1: What is the **Moving thing** that is to the **front left** of the **me**? A1: Car
Q2: Are there any **standing pedestrians** to the **front right** of the **stopped bus**? A2: Yes
Q3: Are there any **standing cars** to the **front right** of the **traffic cone**? A3: No

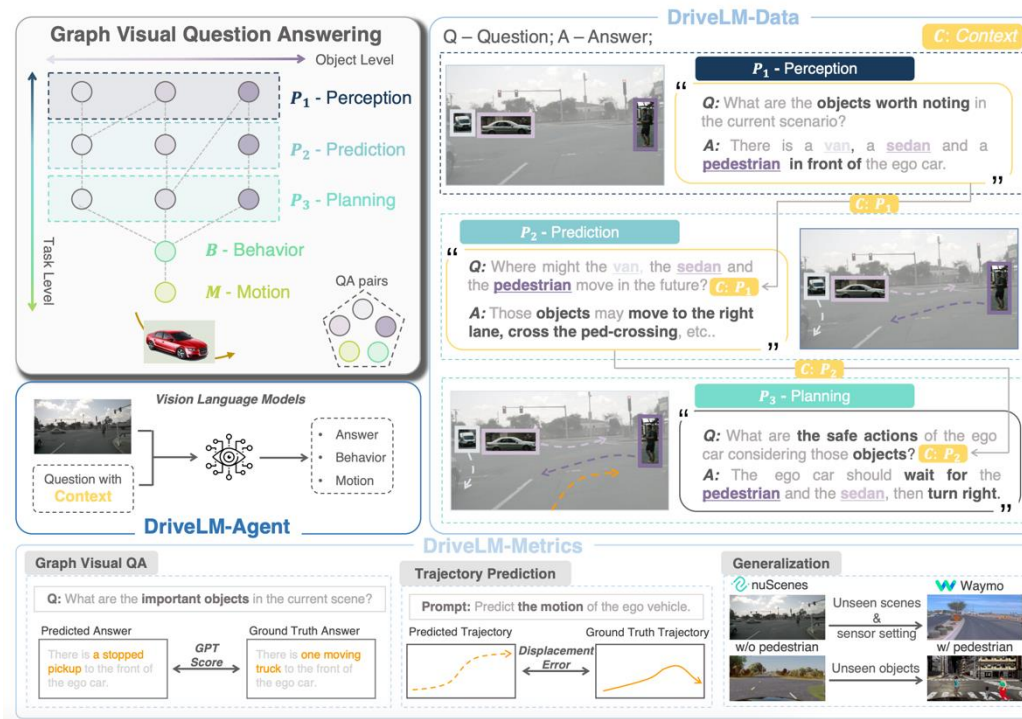
Step ④

Post Processing

- Modify Inappropriate Expressions
the **me** → **me**
standing thing → **standing pedestrian** ...
- Remove Incorrect Combination
standing cars、**moving traffic cones** ...

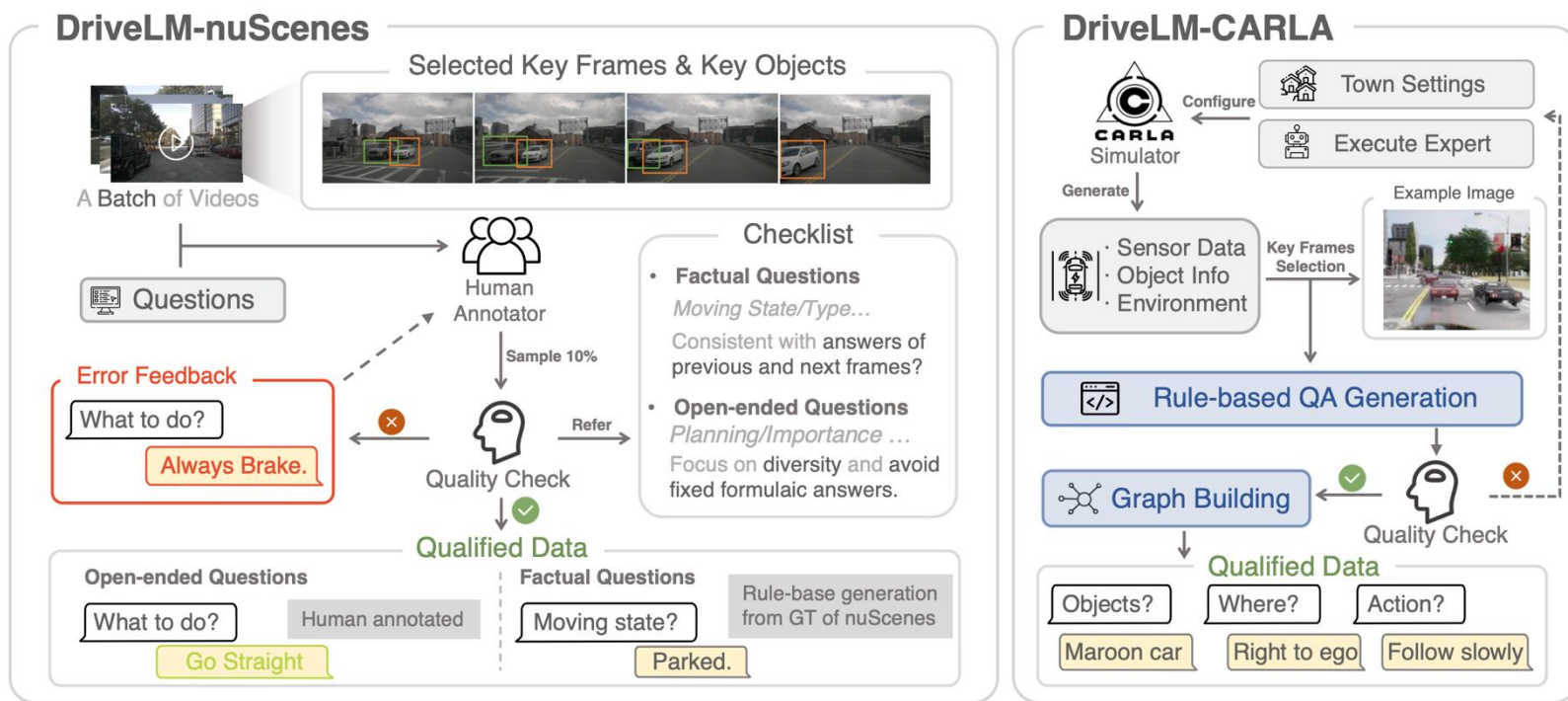
Key Tasks: Specific Domain (Autonomous Driving)

DriverLM- Driving (Sima et al. 2024)



Key Tasks: Specific Domain (Autonomous Driving)

DriverLM- Driving (Sima et al. 2024)



Key Tasks: Specific Domain (Visually-Rich Document)

Form-NLU (Ding et al. 2023)



(a) Digital (D)

(b) Printed (P)

(c) Handwritten (H)

Figure 1: Digital, Printed and Handwritten Form Samples.

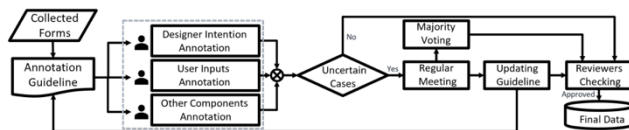


Figure 2: Overall Annotation Workflow.

Task A: Form Layout Analysis

Form 604
Corporations Act 2001
Section 674 (1)

Notice of change of interests of substantial holder

To: Company Name/Scheme: **Terra Resources Limited**

ACN/ARSN: **121 339 704**

1. Details of substantial holder (1)

Name: **Territory Resources Limited**

ACN/ARSN (if applicable): **505 562 116**

There was a change in the interests of the substantial holder on: **18/06/2018**

The previous notice was given to the company on: **18/06/2018**

The previous notice was dated: **18/06/2018**

2. Previous and present voting power

The total number of votes attributable to all the voting shares in the company or voting interests in the scheme that the substantial holder or an associate (2) hold a relevant interest (3) in when last required, and when now required, to give a substantial holding notice to the company or scheme, are as follows:

Class of securities (4)	Previous notice	Present notice	Previous notice	Present notice
	Person's votes	Voting power (5)	Person's votes	Voting power (5)
Ordinary securities	9,280,000	9.28%	13,419,536	7.64%

Categories	Description
Title	Form Title (e.g. "Form 604")
Section	Section Header (e.g. "Details of substantial holder")
Form Key	Form Key Segments (e.g. "Name")
Form Value	Form Value Segments (e.g. "Territory Resources Limited")
Table Key	Table Key Segments (e.g. "Voting Power (5)")
Table Value	Table Value Segments (e.g. "5.29%")
Other	Others (e.g. text paragraphs)

Task B: Key Information Extraction

Form 604
Corporations Act 2001
Section 674 (1)

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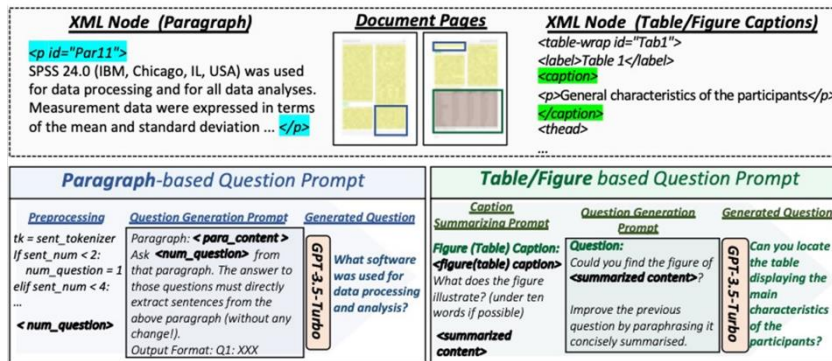
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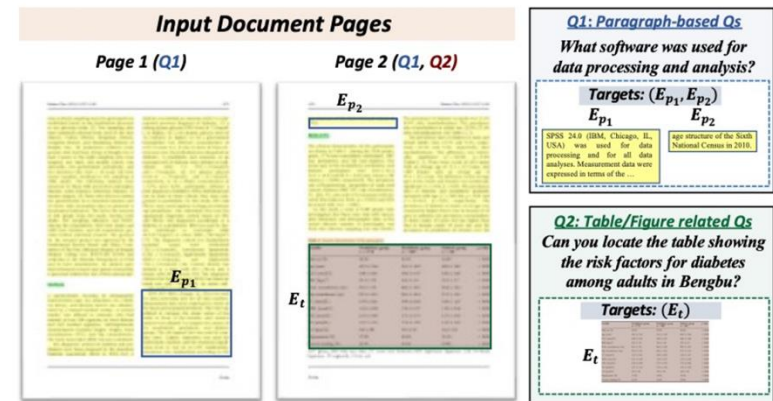
Key	Value
Company Name/Scheme	Tinybeans Group Ltd
ACN/ARSN	168 481 614
Name	Stephen O'Young
There was a change in the interests of the substantial holder on	16/06/2018
The previous notice was given to the company on	26/04/2017
The previous notice was dated	26/04/2017
Class of securities (4)	Ordinary shares
Previous notice Person's votes	6,733,574
Previous notice Voting power (5)	25.75%
Present notice Person's votes	6,733,574
Present notice Voting power (5)	25.75%

Key Tasks: Specific Domain (Visually-Rich Document)

MM-VQA (Ding et al. 2024)



Sample Question generation progress



Tasks of multi-modal cross-page
information retrieval

Datasets for Pretraining

Coarse-grained Image-Text

- Large data volume (from the Internet)
- Short and Noisy (due to the web-crawled nature)



Alt-text: A Pakistani worker helps to clear the debris from the Taj Mahal Hotel November 7, 2005 in Balakot, Pakistan.

Conceptual Captions: a worker helps to clear the debris.



Alt-text: Musician Justin Timberlake performs at the 2017 Pilgrimage Music & Cultural Festival on September 23, 2017 in Franklin, Tennessee.

Conceptual Captions: pop artist performs at the festival in a city.

Conceptual 3M
(Sharma et al. 2018)



<PERSON> was the first US president to attend a tournament in sumo's hallowed Ryogoku Kokugikan arena. (AFP photo)



Hand holding a fresh mangosteen

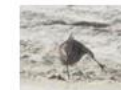


#jellyfish #blue #ocean #pretty Sea Turtle Wallpaper, Aquarius Aesthetic, Blue Aesthetic Pastel, The Adventure Zone, Capricorn And <PERSON>, Life Aquatic, Ocean Life, Jellyfish, Marine Life

Conceptual 12M
(Changpinyo et al. 2020)



Query Image



Not quite sure what the name of this bird is. Saw while walking along the beach in Ocracoke, NC



Young baboon in the campsite at fish river canyon.



This is a boat I saw while walking near the house we rented.



Granite in green glass.

1,000

SBU Captions
(Ordonez et al. 2011)

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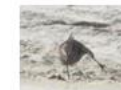


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SBU Captions
(Ordonez et al. 2011)

+) **LAION-5B, -COCO** (Schuhmann et al. Sep 2022): web-crawled images and associated alt-text as captions

Datasets for Pretraining

Coarse-grained Image-Text

- Large data volume (from the Internet)
- Short and Noisy (due to the web-crawled nature)



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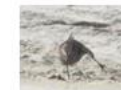


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1,000

SBU Captions
(Ordonez et al. 2011)

- +) **LAION-5B, -COCO** (Schuhmann et al. Sep 2022): web-crawled images and associated alt-text as captions
- +) **COYO-700M** (Byeon et al. Aug 2022): from the CommonCrawl. with several data filtering and cleaning

Datasets for Pretraining

Fine-grained Image-Text

- High-quality through prompting MLLMs
- Longer and Accurate descriptions of the images



COCO: Young children standing on a platform waiting for a train to arrive. Adults and children watching a train slowly leave. A family near a railroad track watching the train pass. People waiting on a platform as a train pulls up. A train station with a green train on the tracks and children waiting for it to go by.

LLaVA: At a train station, a group of people, including both young children and adults, are standing on a platform waiting for a train to arrive. The train is already present on the tracks, partially visible on the right side of the image. Some of the people watch the train closely, while others seem to be patiently anticipating its departure.

There is a total of eight individuals waiting for the train, with one child in the middle of the platform and the others scattered around. A backpack can be found on the far left side of the platform, suggesting that someone may have set it down while waiting.

ShareGPT4V: The image captures a moment at a train station. A green train is on the tracks, moving away from the platform labeled "Platform 2". The train's motion is observed by people standing on the platform, their attention drawn to the departing vehicle.

A red signal stands on the tracks, its vibrant color contrasting with the green of the train. Signs are prominently displayed around the platform. One warns "Beware of Trains", a cautionary reminder of the powerful machines that frequent these tracks. Another sign advises "Please Use The Footbridge To Cross The Line. If You Require Assistance Please Ask The Station Staff", guiding passengers to safely navigate the station.

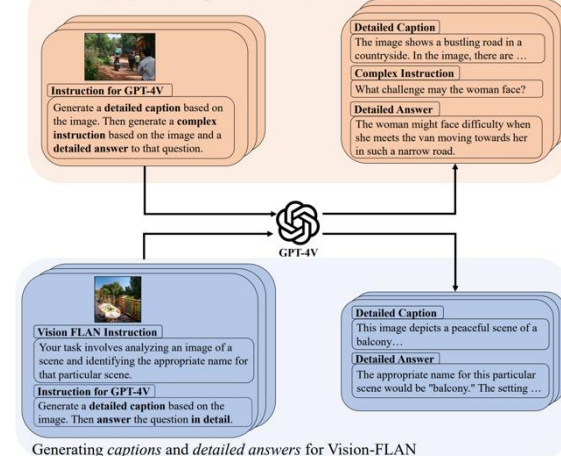
The sky above is a clear blue, providing a serene backdrop to the bustling activity of the station. In the distance, trees can be seen, adding a touch of nature to this man-made setting. The image is a snapshot of everyday life at a train station, capturing both its routine operations and its inherent charm.

(a) Comparison of Captions' Quality

ShareGPT4V

(Chen et al. Nov 2023)

Generating captions, complex instructions and answers for LAION



ALLaVa

(Chen et al. Feb 2024)

Datasets for Pretraining

Video-Text

- Large-scale dataset for the open domain video captioning (10K video clips from 20 categories)
- Each video clip is annotated with 20 English sentences by Amazon Mechanical Turks

MSR-VTT (MicroSoft Research Video-To-Text) (Xu et al. 2016)



1. A black and white horse runs around.
2. A horse galloping through an open field.
3. A horse is running around in green lush grass.
4. There is a horse running on the grassland.
5. A horse is riding in the grass.



1. A man and a woman performing a musical.
2. A teenage couple perform in an amateur musical.
3. Dancers are playing a routine.
4. People are dancing in a musical.
5. Some people are acting and singing for performance.



1. A woman giving speech on news channel.
2. Hillary Clinton gives a speech.
3. Hillary Clinton is making a speech at the conference of mayors.
4. A woman is giving a speech on stage.
5. A lady speak some news on TV.



1. A white car is drifting.
2. Cars racing on a road surrounded by lots of people.
3. Cars are racing down a narrow road.
4. A race car races along a track.
5. A car is drifting in a fast speed.



1. A child is cooking in the kitchen.
2. A girl is putting her finger into a plastic cup containing an egg.
3. Children boil water and get egg whites ready.
4. People make food in a kitchen.
5. A group of people are making food in a kitchen.



1. A player is putting the basketball into the post from distance.
2. The player makes a three-pointer.
3. People are playing basketball.
4. A 3 point shot by someone in a basketball race.
5. A basketball team is playing in front of speculators.

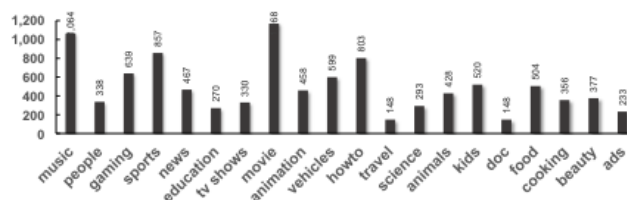
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MSR-VTT
(MicroSoft Research
Video-To-Text)
(Xu et al. 2016)

Dataset	Context	Sentence Source	#Video	#Clip	#Sentence	#Word	Vocabulary	Duration (hrs)
YouCook [5]	cooking	labeled	88	–	2,668	42,457	2,711	2.3
TACos [25, 28]	cooking	AMT workers	123	7,206	18,227	–	–	–
TACos M-L [26]	cooking	AMT workers	185	14,105	52,593	–	–	–
M-VAD [32]	movie	DVS	92	48,986	55,905	519,933	18,269	84.6
MPII-MD [27]	movie	DVS+Script	94	68,337	68,375	653,467	24,549	73.6
MSVD [3]	multi-category	AMT workers	–	1,970	70,028	607,339	13,010	5.3
MSR-VTT-10K	20 categories	AMT workers	7,180	10,000	200,000	1,856,523	29,316	41.2

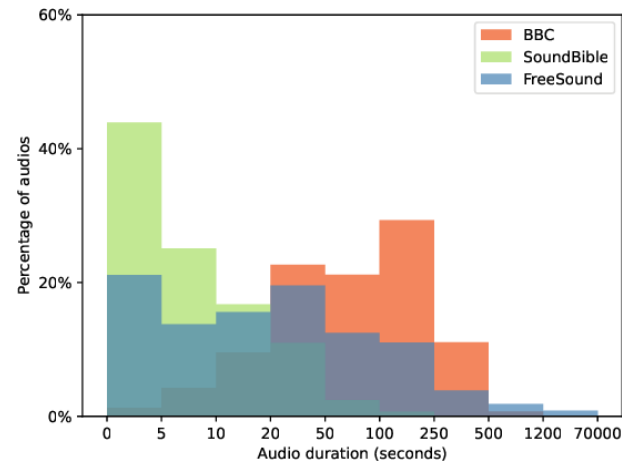


Datasets for Pretraining

Audio-Text

- **Large-scale weakly-labelled audio captioning dataset** (400K audio clips with paired captions)
- Collected from web sources and a sound event dataset (FreeSound, BBC, Soundbible, AudioSet)

WavCaps
Chatgpt-assisted
(Mei et al. 2023)

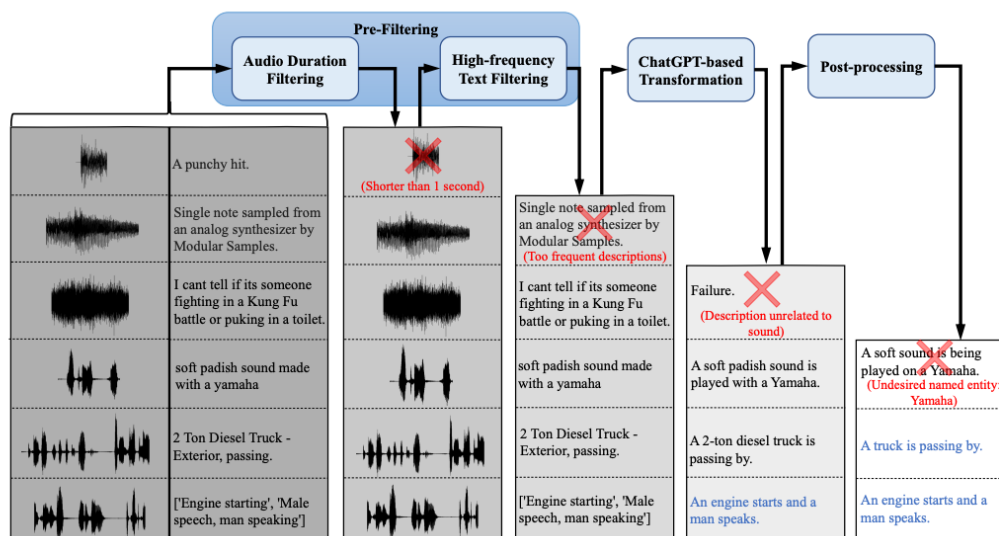


Datasets for Pretraining

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Datasets for Pretraining

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WavCaps Chatgpt-assisted (Mei et al. 2023)

Data sources	Prompts
FreeSound	I will give you a list of descriptions of sounds. Process each individually. Extract the type of the sound and generate an audio caption describing the sound events. The audio caption should be less than 20 words. Delete the author of the sound. Delete locations, city names, country names. Delete the time. Delete device names. Delete the proper noun modifiers, number modifiers, and unit modifiers. Summarize each output into one sentence. Replace all named entities with their hypernyms. Replace people names with "someone". Do not write introductions or explanations. Only describe the sound events and do not use "heard", "recorded". Start each output sentence with its index. Make sure you are using grammatical subject-verb-object sentences. Output "Failure." if the description is not related to sound.
BBC Sound Effects	
SoundBible	
AudioSet SL	I will give you a number of lists containing sound events occurred sequentially in time. Process each individually. Write an one-sentence audio caption to describe these sounds. Make sure you are using grammatical subject-verb-object sentences. Directly describe the sounds and avoid using the word "heard". The caption should be less than 20 words.

Datasets for Pretraining

Audio-Text

- Large-scale weakly-labelled audio captioning dataset (400K audio clips with paired captions)
- Collected from web sources and a sound event dataset (FreeSound, BBC, Soundbible, AudioSet)

WavCaps Chatgpt-assisted (Mei et al. 2023)

Data sources	Raw descriptions	ChatGPT's output
FreeSound	This sound is of a book falling down the staircase in the Stanford University library west stacks. After 2h and 15min. Excerpt.	A book is falling down a staircase. Failure.
BBC Sound Effects	Timber & Wood - Rip saw, carpenters' workshop. Motor Car: Ford Ecsort 1300 (Automatic) - Ford Escort 1300, exterior, passing with horn. (1300cc engine, automatic transmission.)	Someone is using a rip saw in a carpenter's workshop. A car is passing with its horn.
SoundBible	Tasmanian Devil growling screaming hissing. Warning sounds from a Tasmanian Devil in Zoo. Large Tibetan Bells ringing in a temple. Could also use for Monastery or Monks.	An animal is growling, screaming, and hissing. Bells are ringing.
AudioSet SL	['Accelerating, revving, vroom', 'Race car, auto racing'] ['Female speech, woman speaking', 'Whoosh, swoosh, swish']	A race car is accelerating and revving. A woman is speaking while something whooshes.

Challenges of Dataset Creation

1) Data Diversity and Collection Methods (1)

Data Source Complexity

- Where to find the dataset?
- How to gather data from different modalities?

Diverse Representations:

- How to capture a wide range of real-world scenarios access all modalities?
- How to improve generalizability?

Modality Balancing:

- How to avoid the over-representation of any single modality (balancing audio and visual content in an audio-visual dataset)

Challenges of Dataset Creation

1) Data Diversity and Collection Methods (2)

Collection Tools and Technologies

- Diverse collection methods (from web scraping for textual data to using specialized hardware)
- Vision, text, audio collections are relatively easier
- e.g. Bio-sensors, time-series dataset collection tools

Privacy and Ethical Concerns:

- How to manage the sensitive data

Challenges of Dataset Creation

2) Annotation Bottlenecks

Annotation Cost and Time

- Annotation for multimodal data is more time-intensive and cost-intensive
- How to annotate the specialised knowledge across diverse modalities

Training and Hiring Annotators

- How to manage the training to handle the complex or ambiguous multimodal data accurate

Ambiguities in Labelling Across Modalities

- How to make a standardized annotation guidelines
- Less subjective or context-dependent labels

Challenges of Dataset Creation

3) Cross-Modal Alignment

Cross-Modal Synchronization

How to align data from multiple modalities

e.g. aligning text captions with video frames or sensor data with timestamped audio

Annotation Tools for Multimodal Data

What are the advanced annotation platforms capable of handling cross-modal data?
(e.g., synchronizing and labeling text, video, and audio simultaneously)

Lecture 19 & 20: Multimodal Large Language Models and Tunings

1. Introduction and Overview

- Multimodal learning
- Importance and Challenges of integrating multiple modalities (vision, language, sensors, audio)

2. Foundation of Multimodal Large Language Models (MLLMs)

- Multimodal Key Tasks and Benchmark datasets for multimodal tasks
- Challenges in Dataset Creation and Annotation for Multimodal Integration

3. Core Techniques for Multimodal Large Language Models (MLLMs)

- MLLM Architecture
- MM capabilities

4. Multimodal Tunings

- Basic Instruction Tuning Technique to LVLM/MLLM Instruction Tuning
- Multimodal Chain-of-Thought
- Efficient Instruction Tuning Strategies

5. Knowledge Distillation with Multimodality

MLLMs Architecture

Modality-based Encoder

Visual Encoder

ViT (Dosovitskiy et al., 2020),
OpenCLIP (Cherti et al., 2023),
DINOv2 (Oquab et al., 2023),
InternViT (Chen et al., 2023j),
and VCoder (Jain et al., 2023)

... but depends on the task

Audio Encoder

CFormer (Chen et al., 2023b),
HuBERT (Hsu et al., 2021),
BEATS (Chen et al., 2023g),
Whisper (Radford et al., 2023),
and CLAP (Wu et al., 2023e).

... but depends on the task

Unified ImageBind

Unified encoder covering six
modalities, image/video, text, audio,
heatmap, and measurement units, and
depth

Image 

Video 

Audio 

...



Input Projector

MLP

Linear

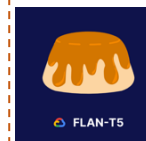
Cross-Attention

Q-Former

P-Former

...

LLM Backbone



...

Output Projector

MLP

Tiny Transformer

...

Modality Generator

Image



Video



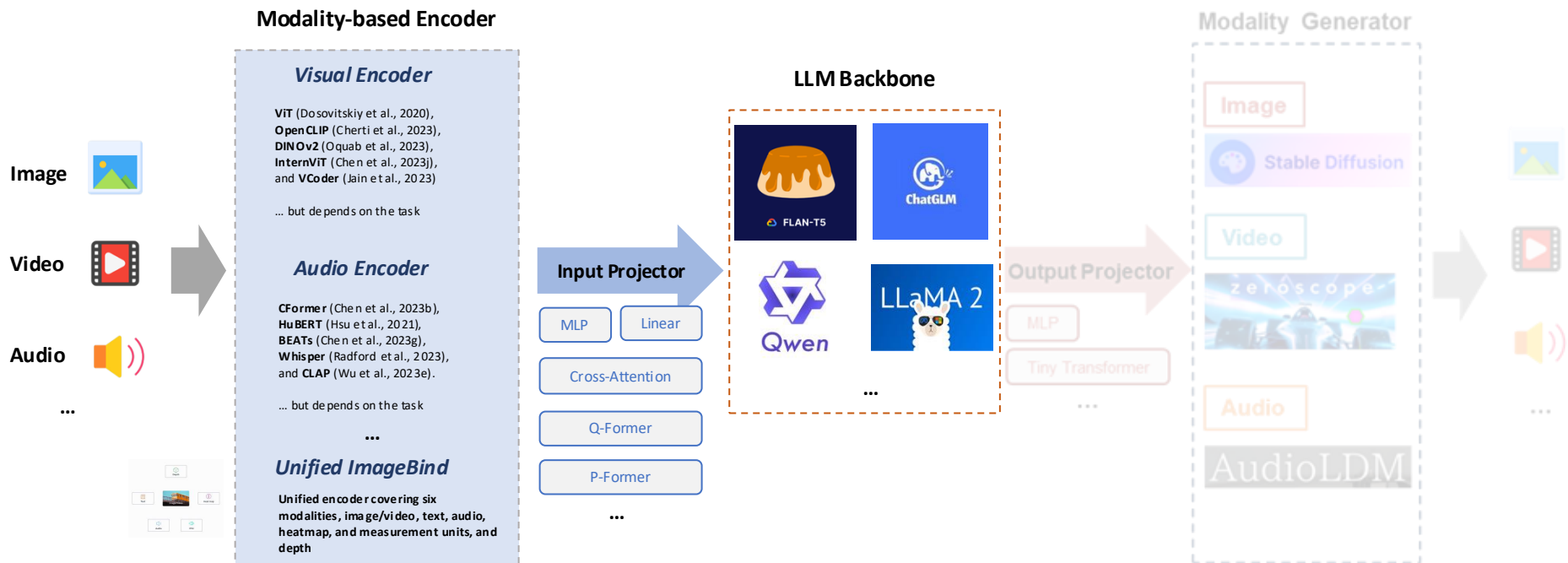
Audio



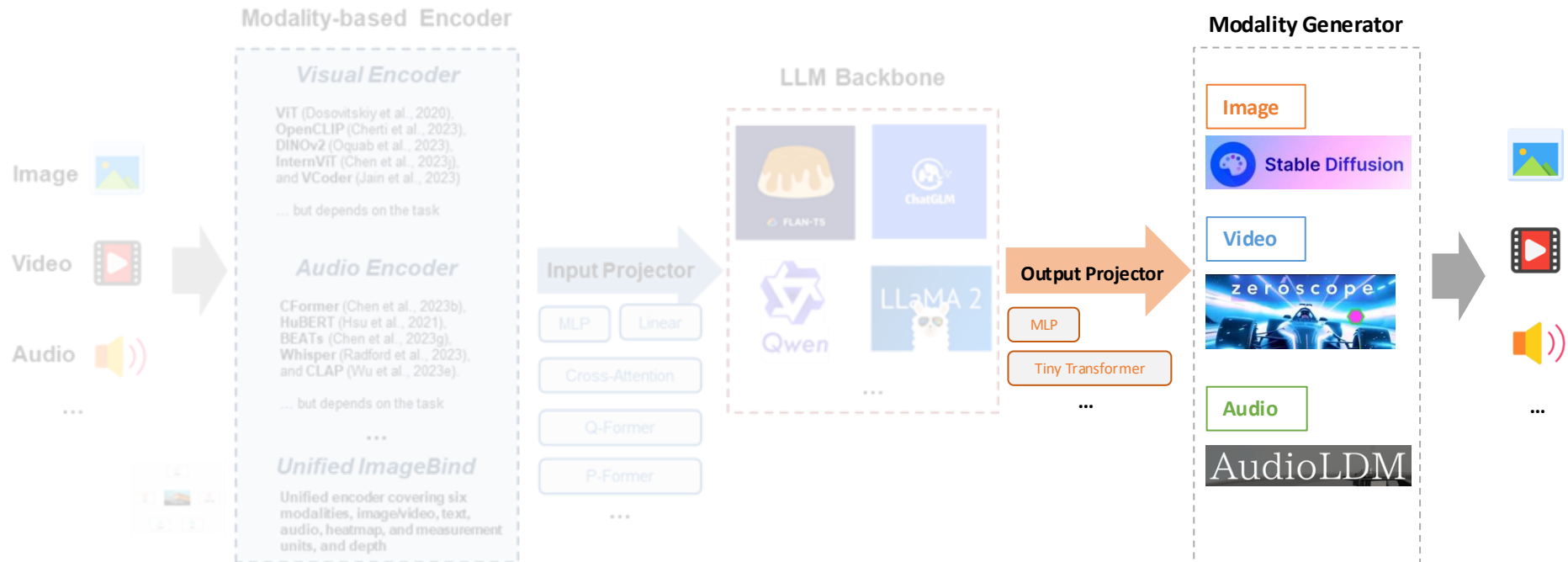
...



MLLMs Architecture – Multimodal Content Understanding

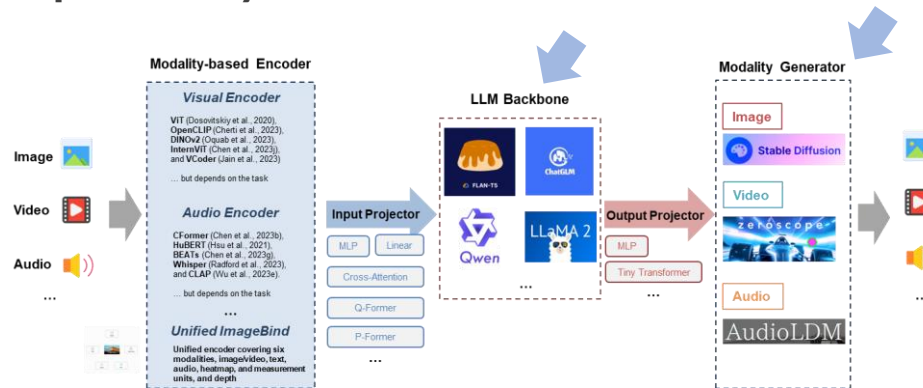


MLLMs Architecture – Multimodality Generation



MLLMs Architecture – Challenges (1)

Dependency on Pretrained Backbones and Frozen Generators

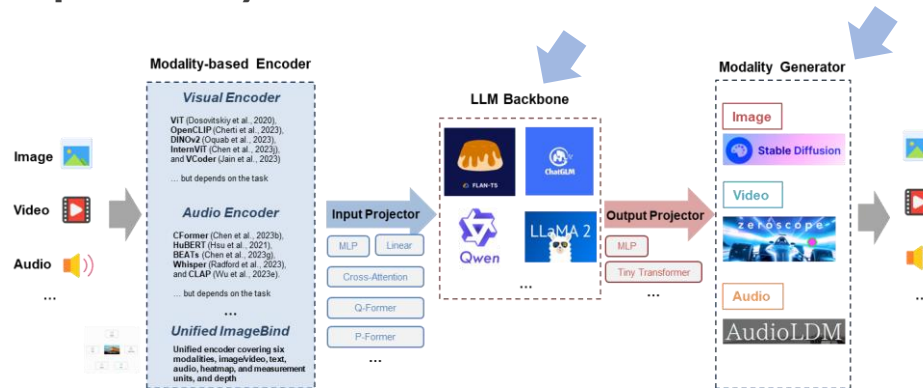


MLLMs maintain their LLM Backbone and Modality Generators in a frozen state

- Their ability to adapt to new, nuanced multimodal interactions is limited
- The fixed configuration means that any updates to the model's understanding of modality interactions require complex workarounds.

MLLMs Architecture – Challenges (1)

Dependency on Pretrained Backbones and Frozen Generators



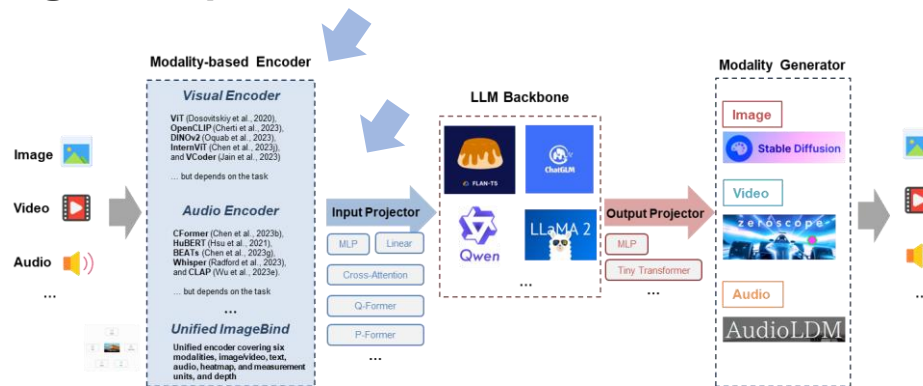
MLLMs maintain their LLM Backbone and Modality Generators in a frozen state

- Their ability to adapt to new, nuanced multimodal interactions is limited
- The fixed configuration means that any updates to the model's understanding of modality interactions require complex workarounds.

Potential Solution: We may develop **Adaptable Backbone Architectures** that allow selective fine-tuning or partial unfreezing of specific layers within the LLM Backbone or Modality Generators. (e.g. PEFT or LoRA)

MLLMs Architecture – Challenges (2)

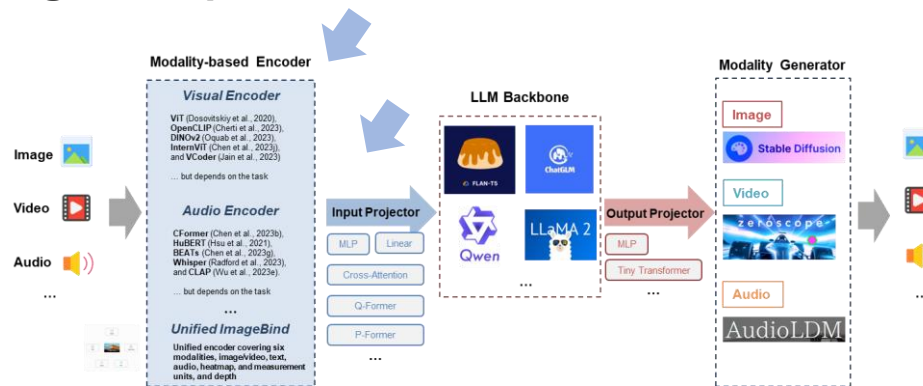
High Computational Costs



Processing diverse modalities through separate Encoders and Projectors before unifying the information in the LLM Backbone requires significant computational resources, especially with large-scale data

MLLMs Architecture – Challenges (2)

High Computational Costs

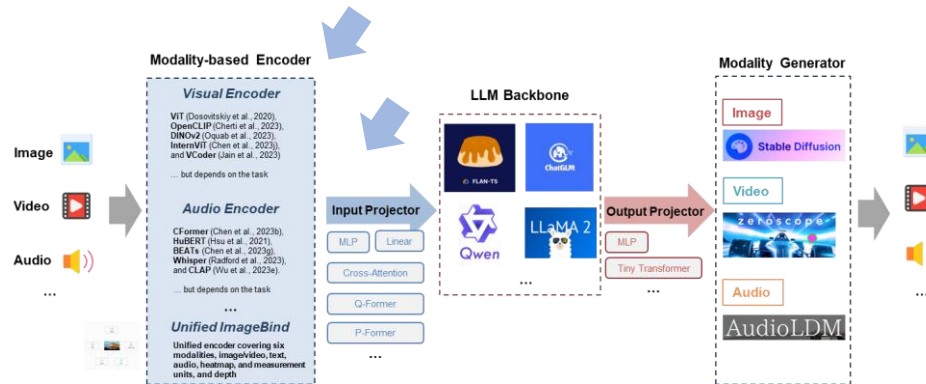


Processing diverse modalities through separate Encoders and Projectors before unifying the information in the LLM Backbone requires significant computational resources, especially with large-scale data

Potential Solution: Mixture-of-Experts (MoE) techniques can be used to activate only relevant parts of the model for specific tasks or modalities, thus lowering resource demands.

MLLMs Architecture – Challenges (3)

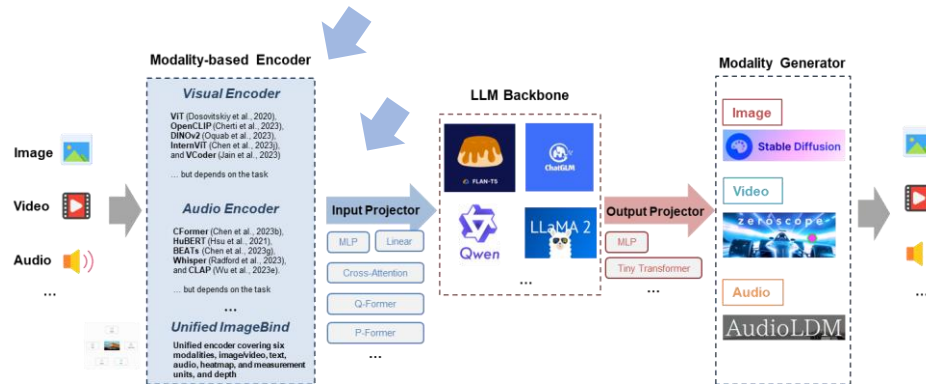
Limited Flexibility with New or Evolving Modalities



The static design of Input Projectors and Modality Encoders for specific modalities **restricts the system's ability to integrate new or uncommon modalities** (e.g., biosignals, environmental data)

MLLMs Architecture – Challenges (3)

Limited Flexibility with New or Evolving Modalities

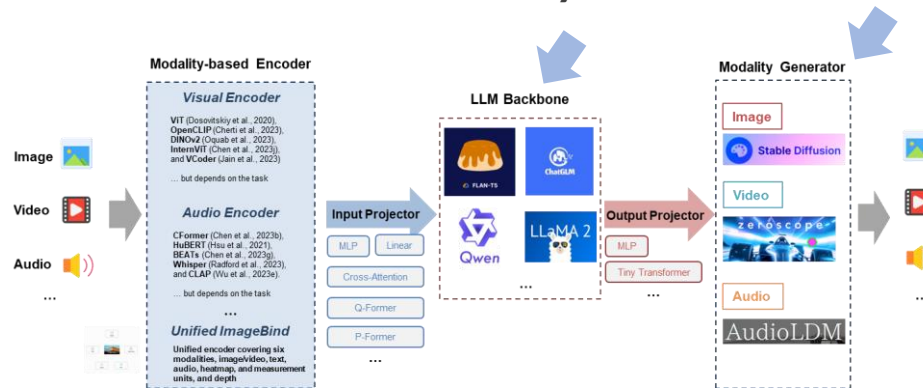


The static design of Input Projectors and Modality Encoders for specific modalities **restricts the system's ability to integrate new or uncommon modalities** (e.g., biosignals, environmental data)

Potential Solution: Develop a Plug-and-Play Modality Integration Framework where dynamic modality adapters or universal imagebind encoders can process new or evolving data types without retraining the entire model.

MLLMs Architecture – Challenges (4)

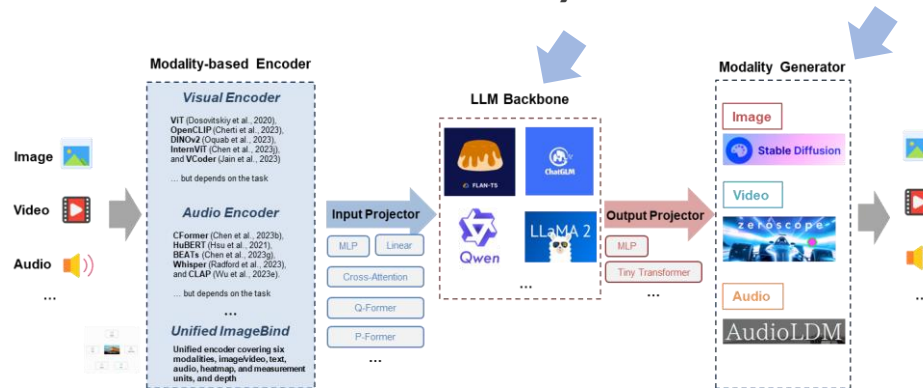
Evaluation and Consistency Across Modalities



- **The evaluation metrics and methods** for assessing multimodal tasks are not yet standardized, especially for models using frozen backbones and generators.
- **Maintaining consistency in how different modalities contribute to the output** can be challenging, particularly in dynamic scenarios.

MLLMs Architecture – Challenges (4)

Evaluation and Consistency Across Modalities



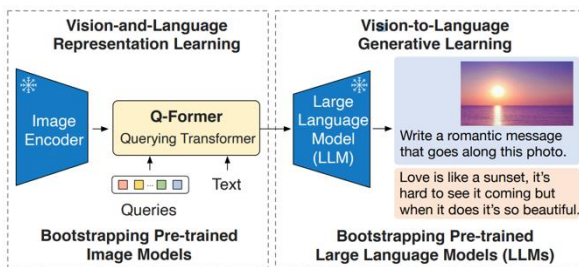
- **The evaluation metrics and methods** for assessing multimodal tasks are not yet standardized, especially for models using frozen backbones and generators.
- **Maintaining consistency in how different modalities contribute to the output** can be challenging, particularly in dynamic scenarios.

Potential Solution: Evaluation Metrics for Multimodal Tasks by developing metrics that consider modality-specific contributions, such as Multimodal BLEU or Multimodal F1 Scores

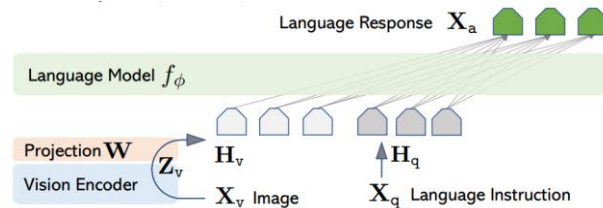
MM Capabilities – Content Understanding and Text Generation

Image-text understanding

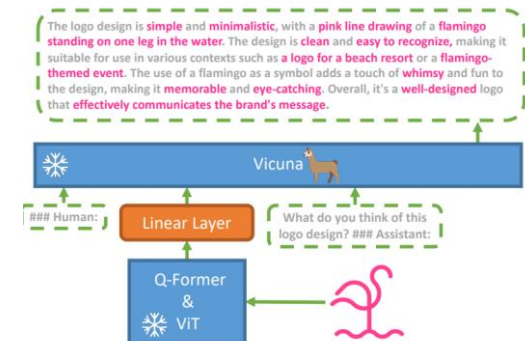
BLIP-2 (Li et al. 2023)



LLaVa (Liu et al. 2023)



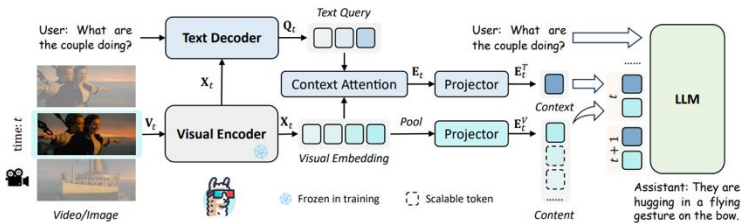
MiniGPT4 (Zhu et al. 2023)



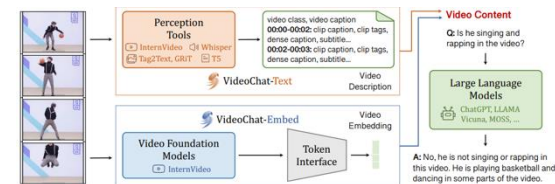
MM Capabilities – Content Understanding and Text Generation

video-text understanding & audio-text understanding

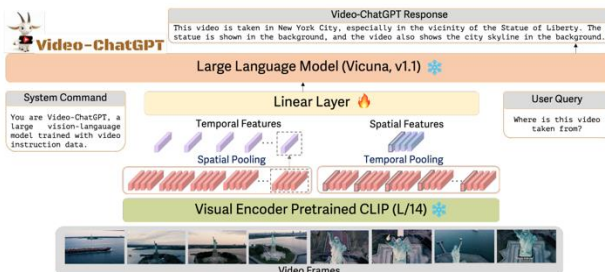
LLaMa-VID (Li et al. 2023)



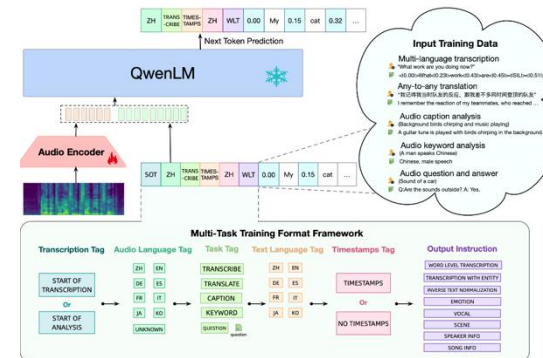
VideoChat (Li et al. 2024)



Video-ChatGPT (Maaz et al. 2023)



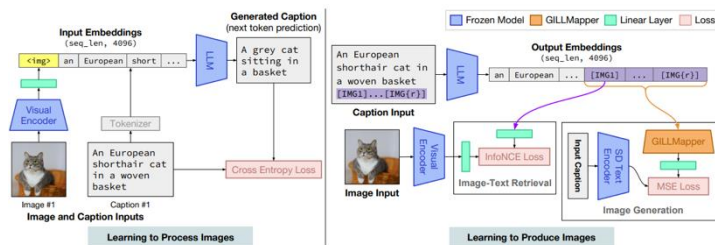
QwenAudio (Chu et al. 2023)



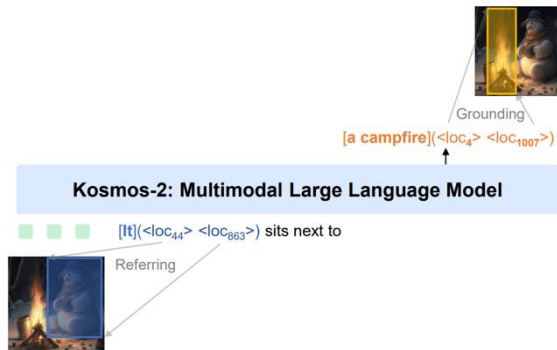
MM Capabilities – Grounding and Referring

image-text output

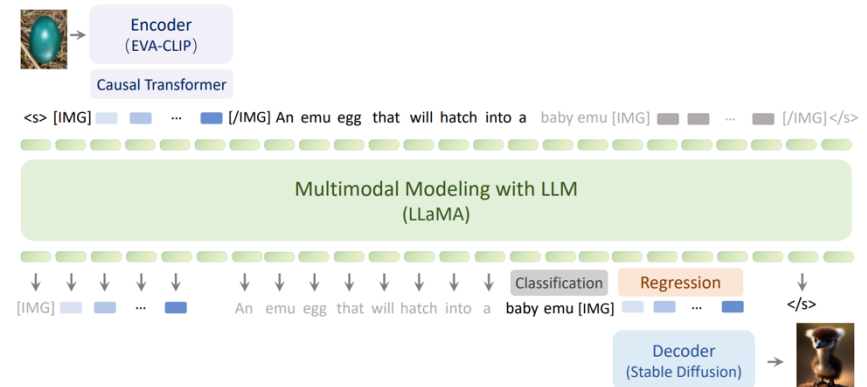
GILL (Koh et al. 2023)



Kosmos-2 (Koh et al. 2023)



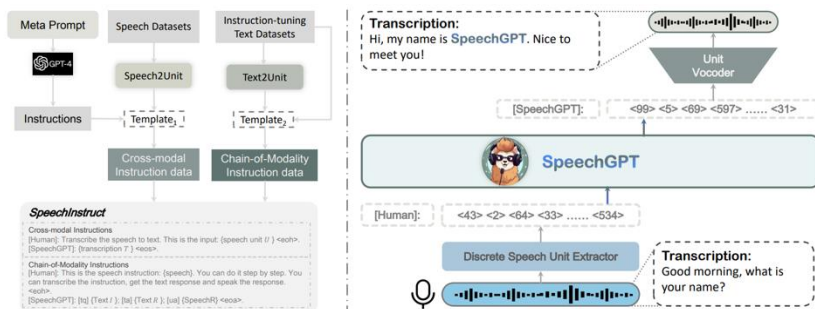
Emu (Sun et al. 2024)



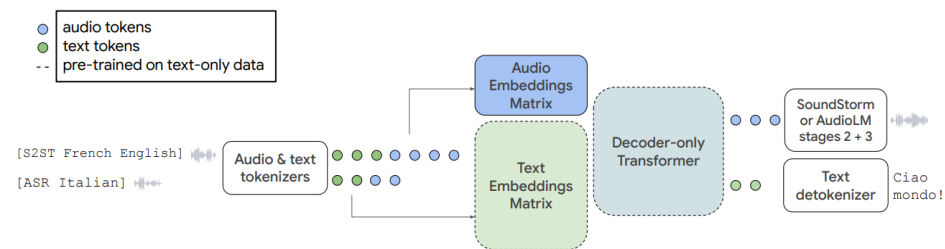
MM Capabilities – Grounding and Referring

speech/audio-text output

SpeechGPT (Zhang et al. 2023)



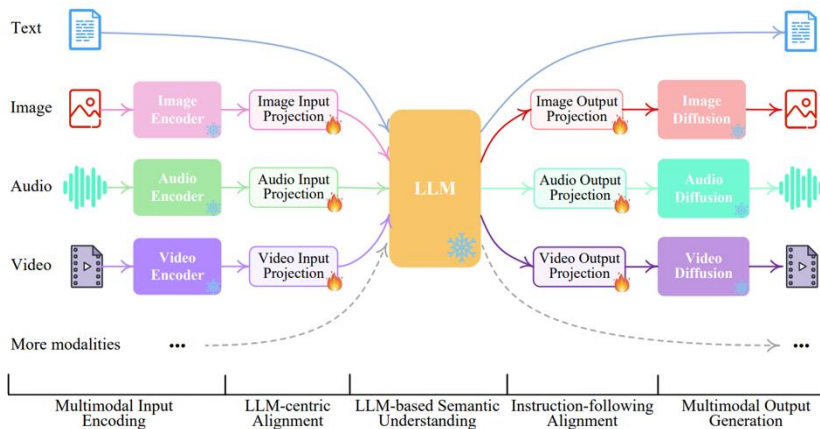
AudioPaLM (Rubenstein et al. 2024)



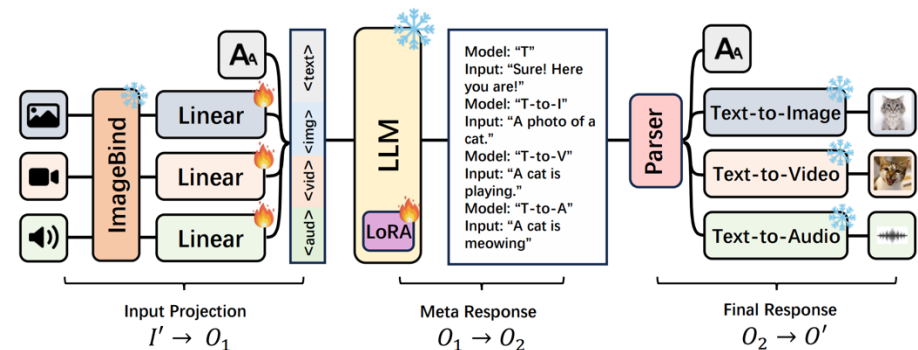
MM Capabilities – Any-to-Any Modalities

End-to-End MLLMs

NEXT-GPT (Wu et al. 2023)



ModaVerse (Wang et al. 2024)



Lecture 19 & 20: Multimodal Large Language Models and Tunings

1. Introduction and Overview

- Multimodal learning
- Importance and Challenges of integrating multiple modalities (vision, language, sensors, audio)

2. Foundation of Multimodal Large Language Models (MLLMs)

- Multimodal Key Tasks and Benchmark datasets for multimodal tasks
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3. Core Techniques for Multimodal Large Language Models (MLLMs)

- MLLM Architecture
- MM capabilities

4. Multimodal Tunings

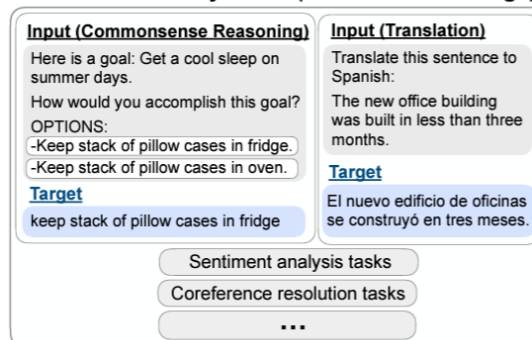
- Basic Instruction Tuning Technique to LVLM/MLLM Instruction Tuning
- Multimodal Chain-of-Thought
- Efficient Instruction Tuning Strategies

5. Knowledge Distillation with Multimodality

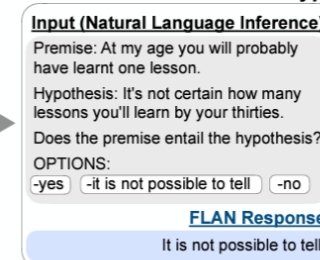
Instruction Tuning

- An approach to train a pretrained model to perform specific tasks based on the provided task descriptions which outline how a task should be performed (i.e. instructions)
- Why do we need Instruction Tuning?
 - Cross-task generalizability is bad
 - Previous framework cannot generalize well to unseen tasks
 - Aim: model can understand human instructions describing how a specific task should be performed and solve unseen tasks based on that, and thus improves the zero-shot generalization ability.

Finetune on many tasks ("instruction-tuning")

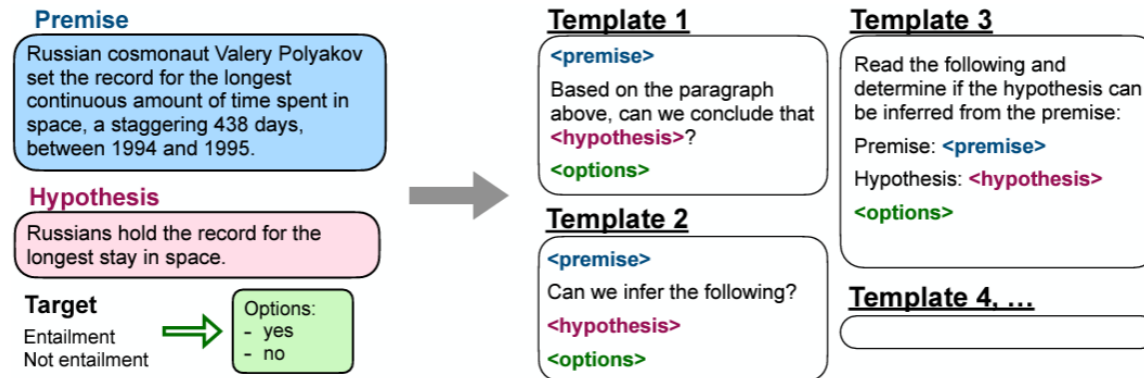


Inference on unseen task type



Instruction Tuning

- *How it works*
 - Mixture of **<Instruction, Input, Target>** from diverse tasks, with diverse templates to describe each type of task as the instruction.
 - Tune the model by giving **< Instruction, Input >** as the input to generate **Target**



LVLM Instruction Tuning

*How does instruction tuning works for **Language + Vision (Image)***

- *Freeze main LLM/Vision Foundation Models and finetune some layers/modules only.*
- *Self-construct a large vision-language instruction tuning corpus based on current vision-language downstream task benchmarks*

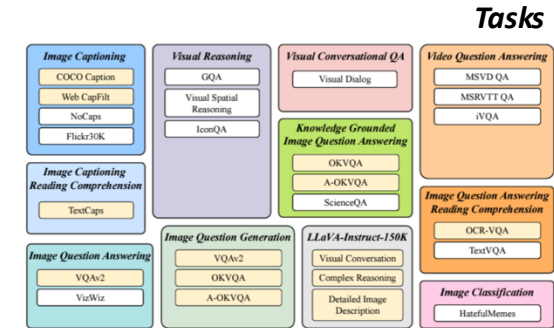


LVLM Instruction Tuning

InstructBLIP

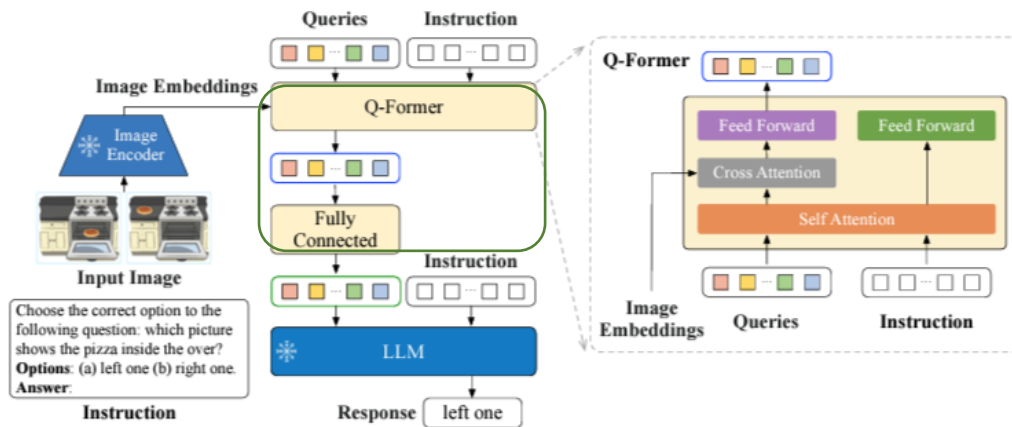
Finetune bridging module (Q-Former) for language and image

- Instruction tasks: QA, image captioning, visual reasoning, etc.
- Multiple predefined instruction templates for each task



Instruction examples

Task	Instruction Template
IC	<Image>Write a short description for the image.
	<Image>A short image caption:
	<Image>A photo of
VQA	<Image>{Question}
	<Image>What is the answer to the following question? "{Question}"
	<Image>Based on the image, respond to this question: {Question}
VQG	<Image>Given the image, generate a question whose answer is: {Answer}. Question:
	<Image>Based on the image, provide a question with the answer: {Answer}. Question:
	<Image>Considering the picture, come up with a question where the answer is: {Answer}.

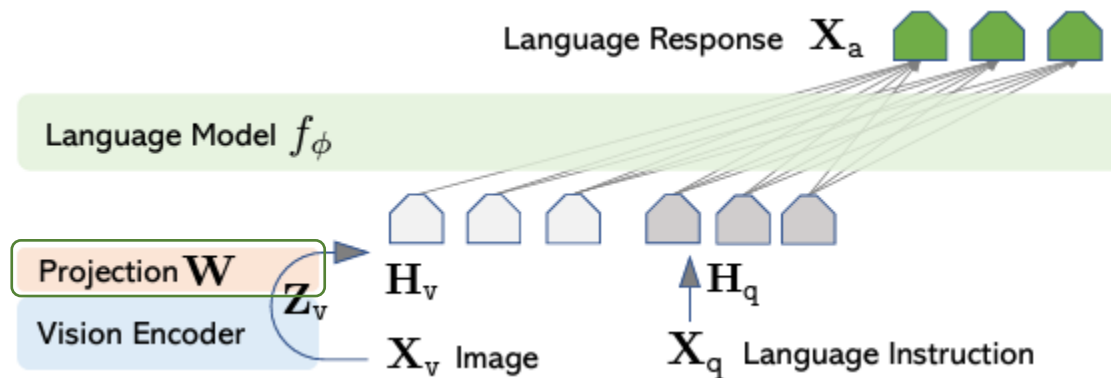


LVLM Instruction Tuning

LLaVA

Finetune image projection layer

- Instruction tasks: QA, image captioning, visual reasoning (158k instances)



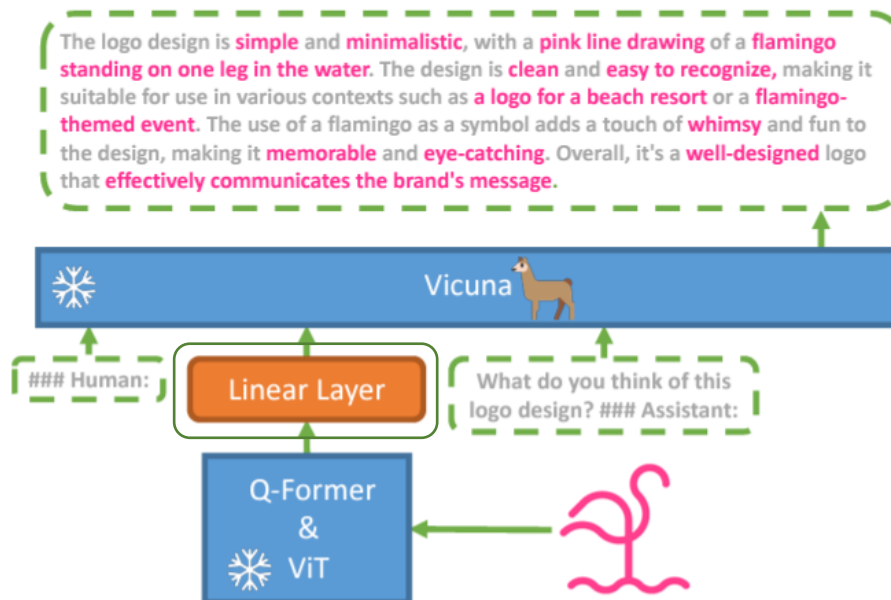
Instruction examples

Task	Instruction Template/ Examples
IC	<Image>Describe the following image in detail
	<Image>Summarize the visual content of the image
	<Image>Share a comprehensive rundown of the presented image
VQA	<Image>{Question}
VR	<Image>What does the image suggest about the winter season in the city?
	<Image>What challenges do these people face?

LVLM Instruction Tuning

MiniGPT4

Finetune image projection layer



Instruction examples

Task	Instruction Template/Examples
IC	<Image>Describe the following image in detail
	<Image>Describe this image in detail
VQA	<Image>{Question}
VR	<Image>Explain why this meme is funny
	<Image>What challenges do these people face?
Other	<Image> What is the issue with my plant? What should I do?
	<Image> Can you give a short introduction to this movie?
	<Image> Can you write me a beautiful poem about this image?
	<Image> Write brief HTML/JS to turn this mock-up into a colorful website.

Chain-of-Thought (CoT)

A prompting technique to invoke the generation of reasoning procedure for LLM

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Multimodal Chain-of-Thought (CoT)

A tuning technique to introduce reasoning for MLLM through rationale generation

An example of detailed reasoning steps in GQA dataset

Question: What appliance is to the right of the cabinet?

###

Please think step by step and provide the bounding box coordinate of the region that can help you answer the question better.

###

Reasoning steps: 1. Identify the cabinet in the image.
2. Observe the area to the right of the identified cabinet.
3. Look for any appliance located to the right side of the cabinet.
4. Determine the name of the appliance found in this location

CoT BBox: [163, 44, 206, 67]

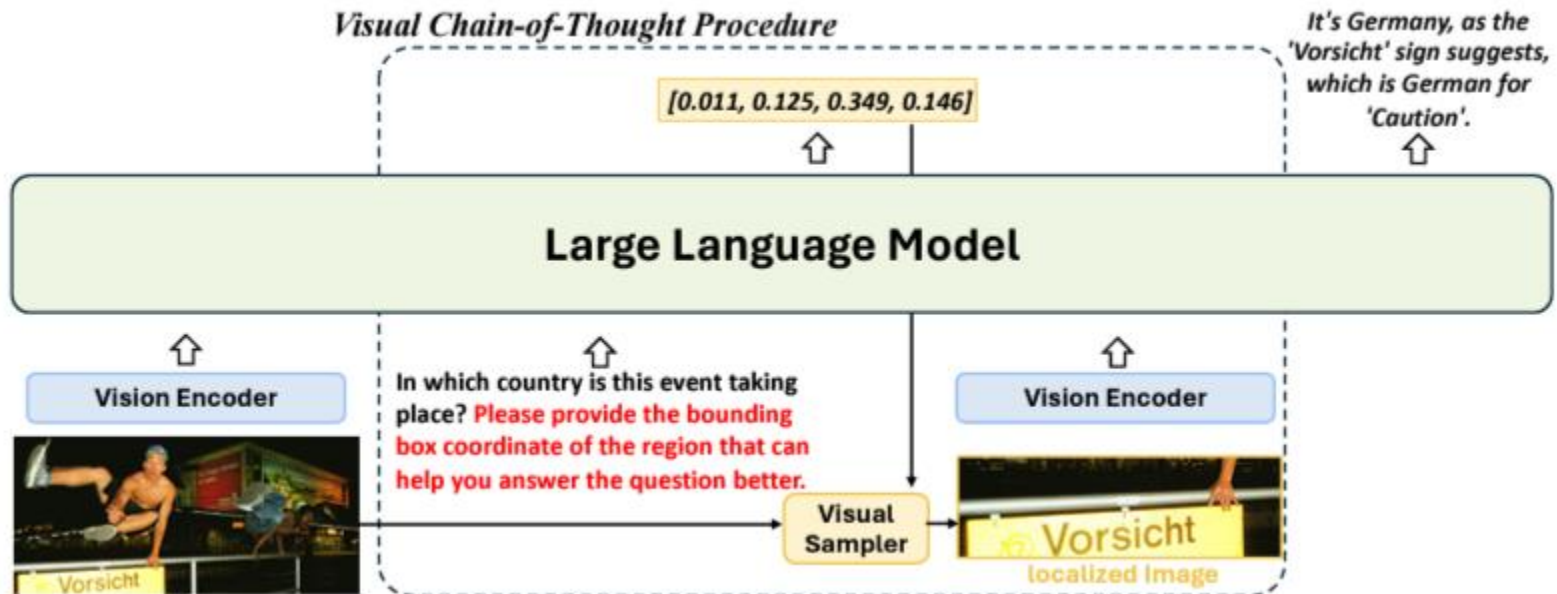
Answer

The appliance is a microwave.



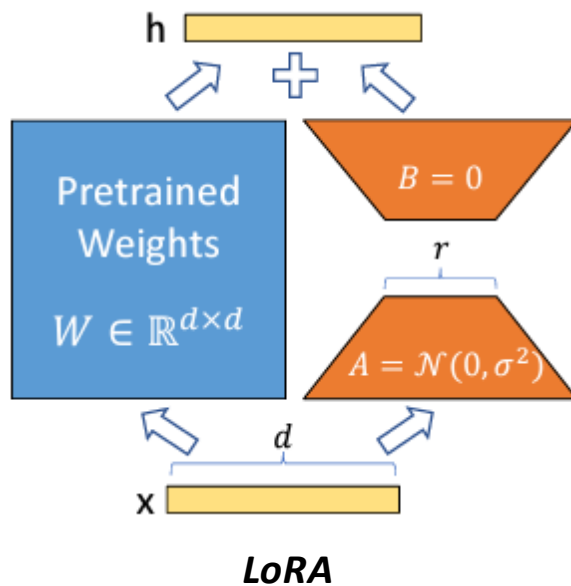
Multimodal Chain-of-Thought (CoT)

VisCoT



Parameter-Efficient Tuning Techniques

LoRA and its variants

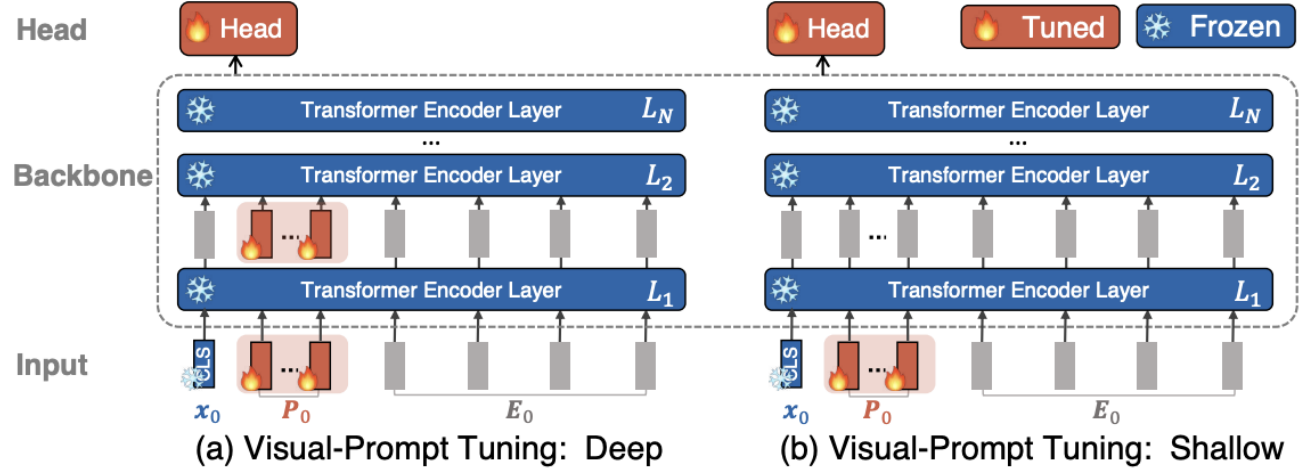
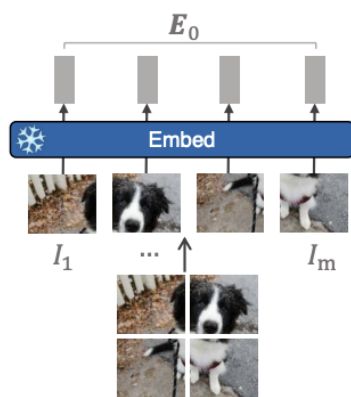


Low-Rank Adaptation (LoRA)

- Reparameterization
- Train A and B only instead of W
- Updated weights $W' = W + AB$
- Do not introduce additional inference latency
- Up to 2/3 reduction in VRAM usage, 25% speed up in training

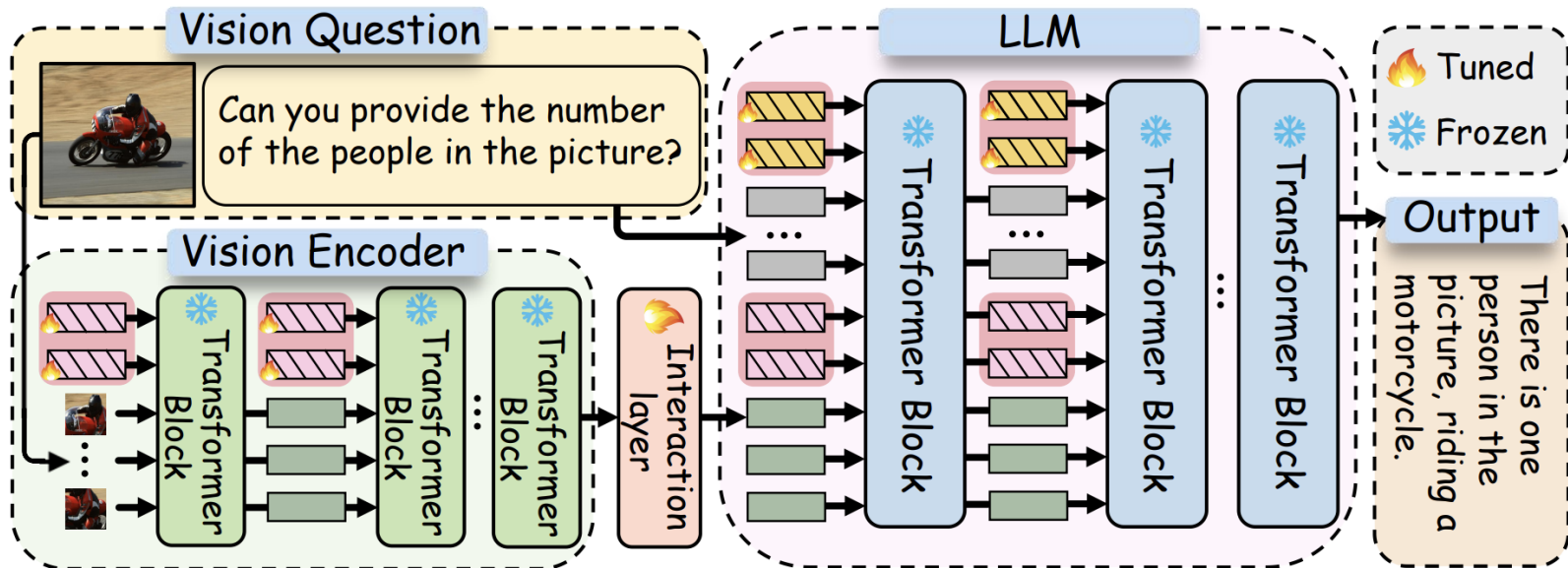
Parameter-Efficient Tuning Techniques

Prompt Tuning



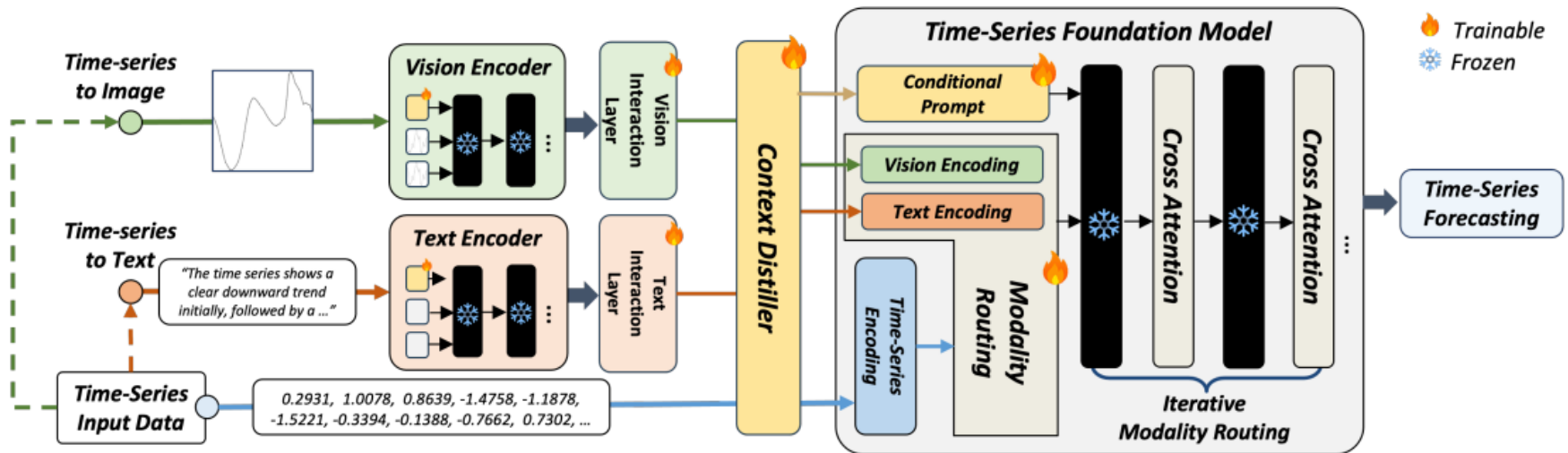
Parameter-Efficient Tuning Techniques

M2PT (2024)



Parameter-Efficient Tuning Techniques

UniCast (2026)



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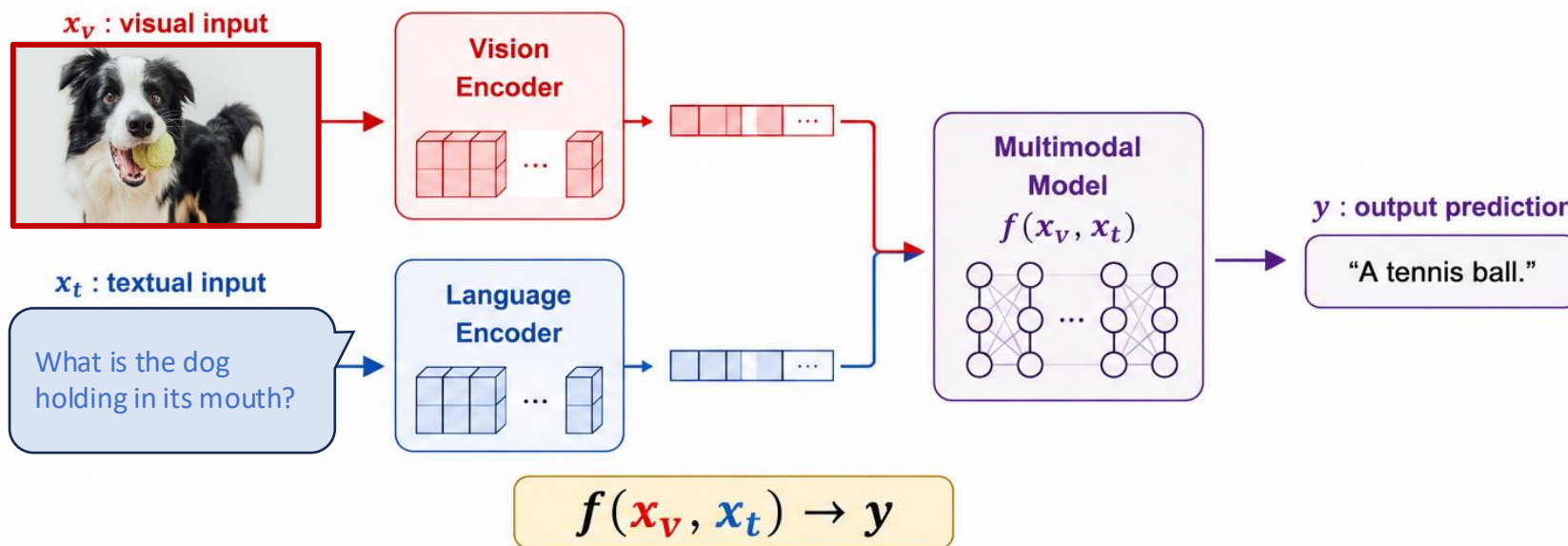
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5. Knowledge Distillation with Multimodality

5 Knowledge Distillation with Multimodality

Modern Multimodal Large Language Models (MLLMs) are extremely powerful, but they are also extremely expensive.



where:

- x_v : visual input
- x_t : textual input
- $f(\cdot)$: multimodal model
- y : output prediction

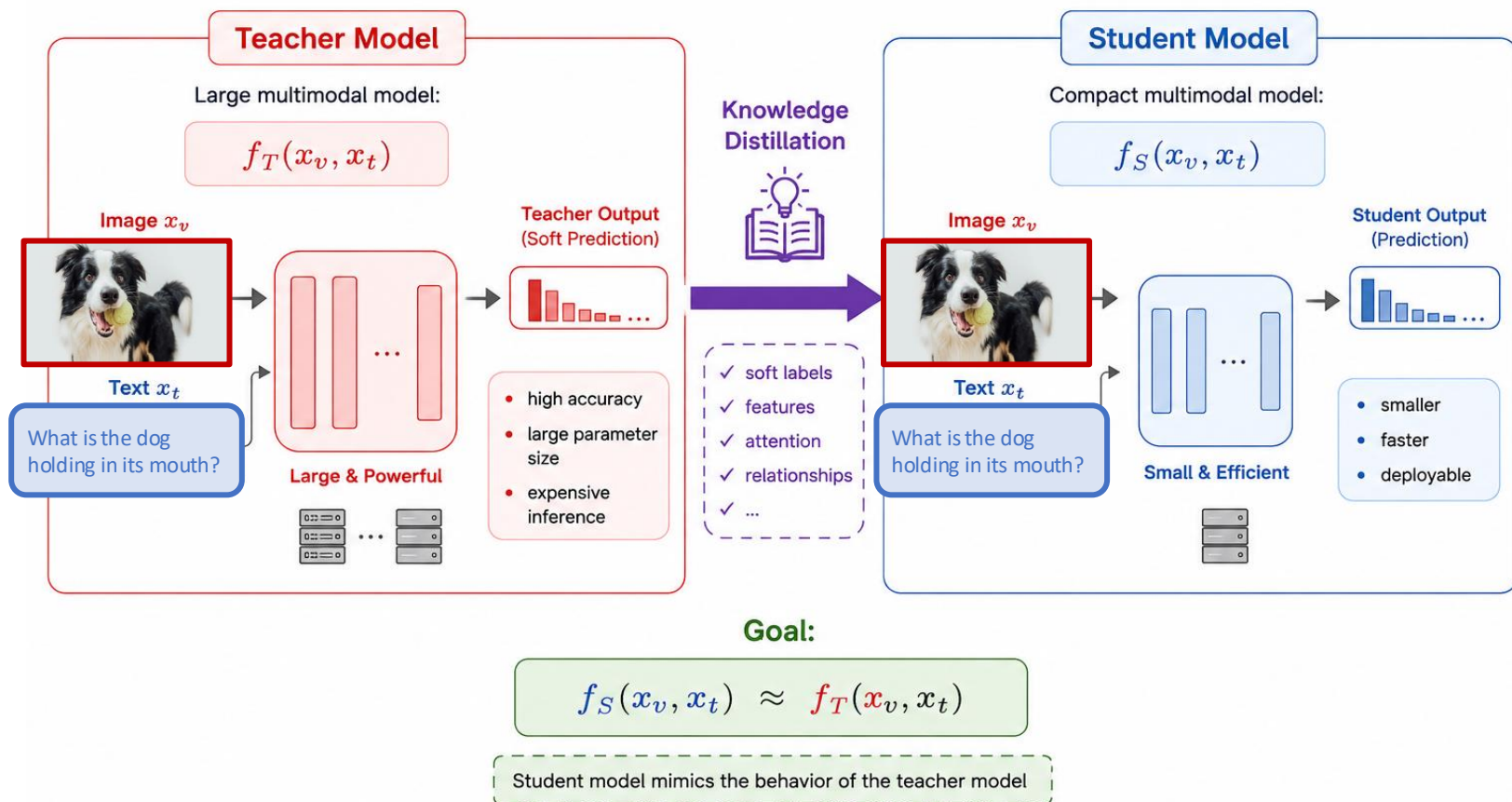
As the model size increases



Small Model $\approx$ Large Model

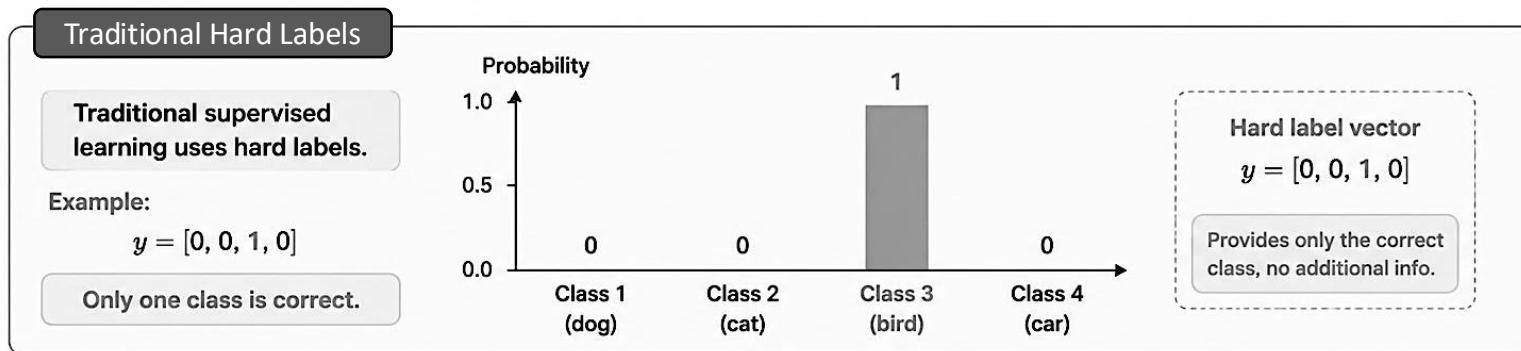
5 Knowledge Distillation with Multimodality

Knowledge Distillation transfers knowledge from a large model (teacher) to a smaller model (student).

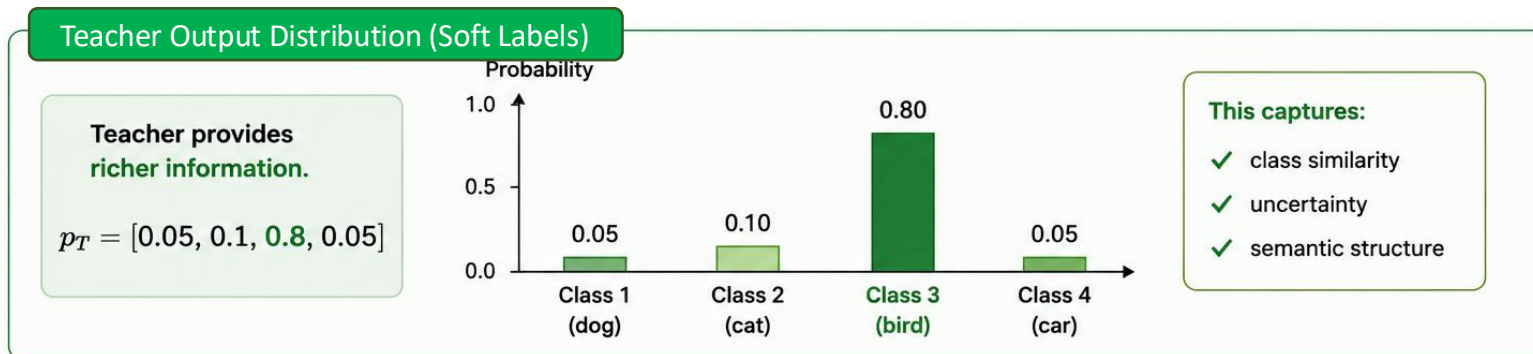


5 Knowledge Distillation with Multimodality

Traditional supervised learning uses hard labels.



Teacher Output Distribution (Soft labels contains richer information than hard labels)



Student learns:



$$p_S \approx p_T$$

Student learns the full probability distribution from the teacher.



5 Knowledge Distillation with Multimodality

Temperature Scaling: Distillation commonly uses temperature scaling.

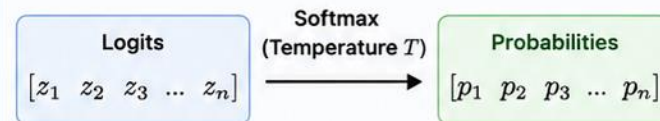
Softmax with Temperature

Softmax:

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

where:

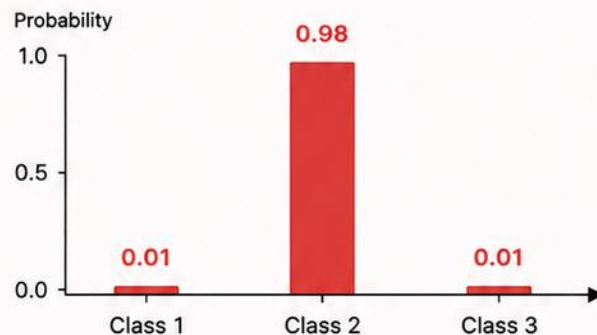
- z_i : logit
- T : temperature



Effect of Temperature

Small T (e.g., $T = 0.1$)

Sharpened distribution (more confident)



$[0.01, 0.98, 0.01]$

Large T (e.g., $T = 5$)

Smoothed distribution (less confident)



$[0.2, 0.5, 0.2]$

5 Knowledge Distillation with Multimodality

Temperature Scaling: Distillation commonly uses temperature scaling.

Softmax with Temperature

Softmax:

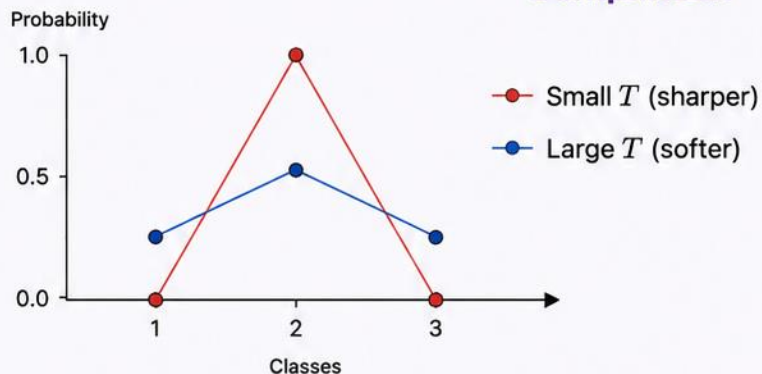
$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

where:

- z_i : logit
- T : temperature

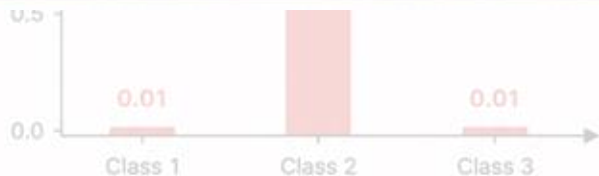


Comparison



**Soft targets
contain more
information.**

- ✓ Capture inter-class relationships
- ✓ Reflect uncertainty
- ✓ Provide richer supervision



[0.01, 0.98, 0.01]

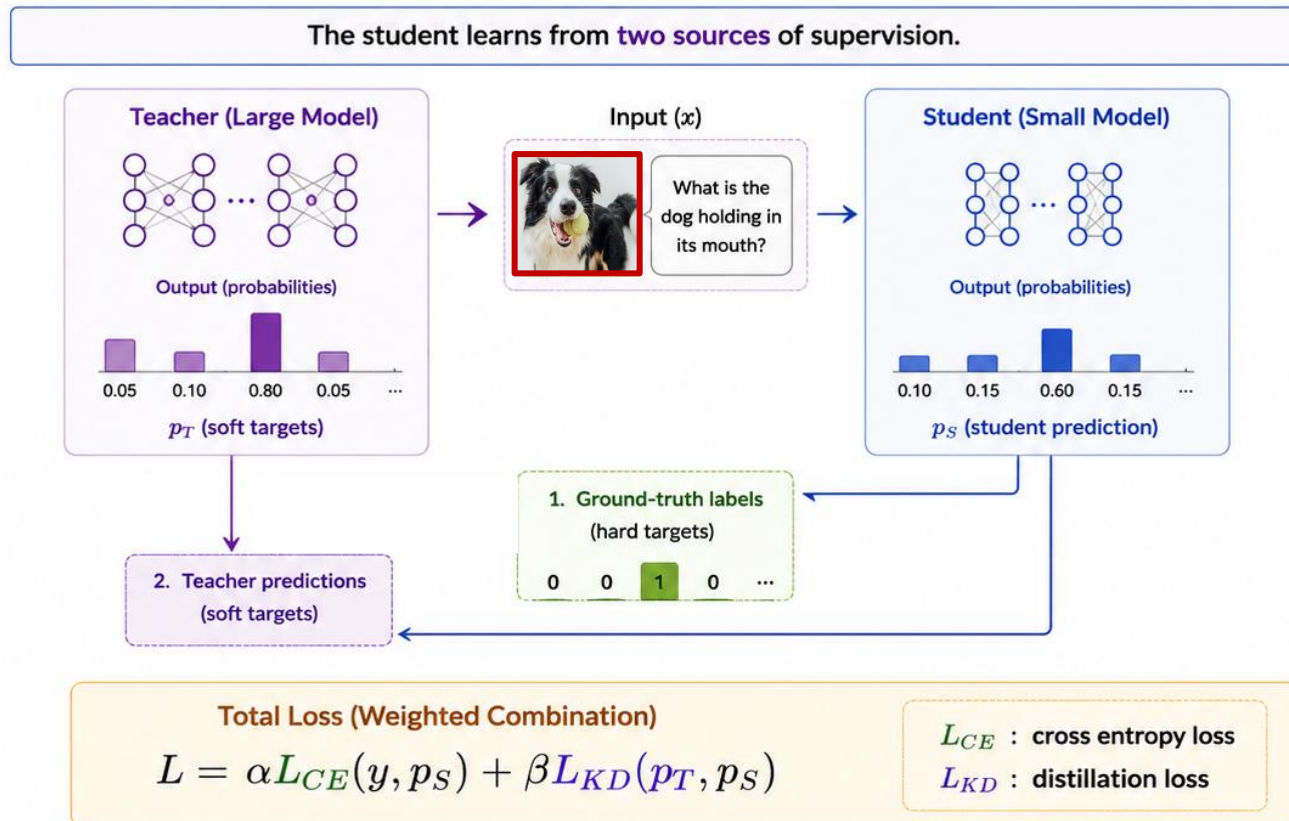


[0.2, 0.5, 0.2]

5 Knowledge Distillation with Multimodality

Knowledge Distillation Loss

The student learns from two sources of supervision.



Knowledge Distillation Loss

The student aims to mimic the teacher as closely as possible.

KL Divergence Distillation

Most commonly, the distillation loss is KL divergence.

KL Distillation Loss

$$L_{KD} = KL(p_T \| p_S)$$

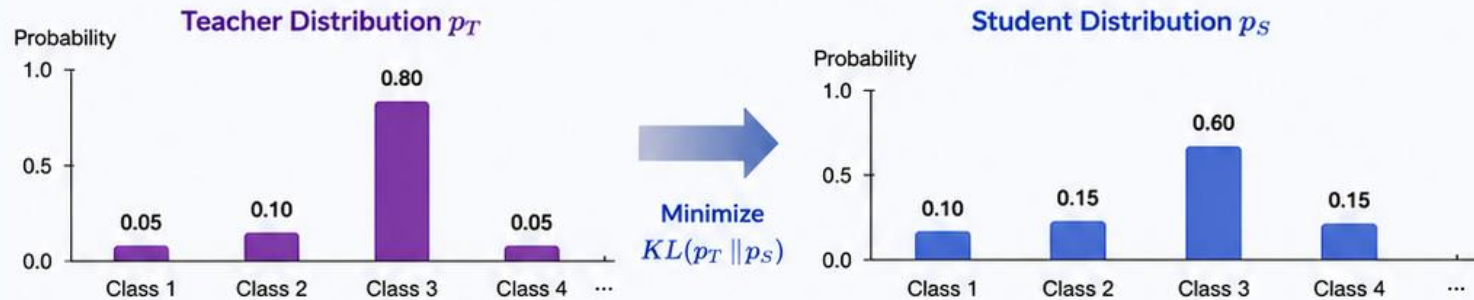
Expanded Form

$$KL(p_T \| p_S) = \sum_i p_T(i) \log \frac{p_T(i)}{p_S(i)}$$

Intuition

KL divergence penalises differences between student and teacher probability distributions, weighted by the teacher's confidence.

Goal: make student distribution close to teacher distribution



Distillation becomes harder in multimodal systems because knowledge exists across multiple modalities.

We must transfer:



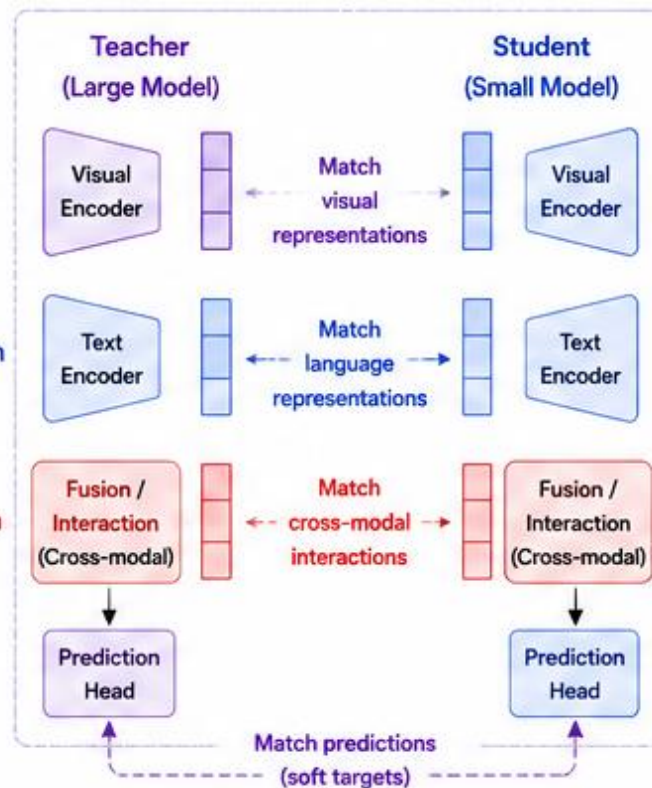
Visual representation



Language representation



Cross-modal interaction



Feature distillation

Student matches intermediate representations.

$$L_{feat} = \|h_T - h_S\|_2^2$$

h_T :teacher feature
 h_S :student feature

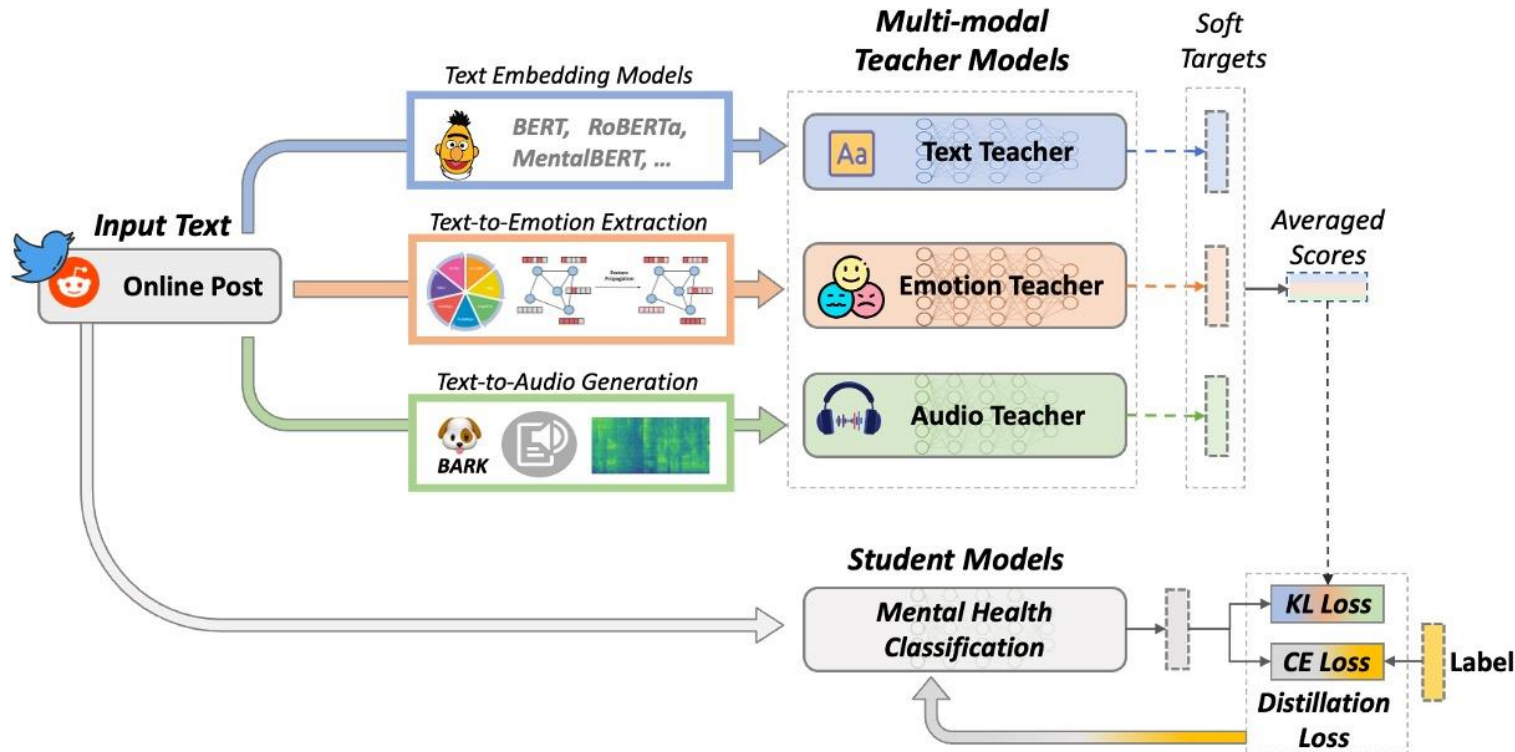
Attention distillation

Transformer-based MLLMs contain attention maps.

$$L_{attn} = \|A_T - A_S\|_2^2$$

A_T :teacher attention
 A_S :student attention

Multimodal Multi-Teacher Knowledge Distillation (CIKM 2024)



Multimodal and Multi-Teacher Knowledge Distillation model for Mental Health Classification, leveraging insights from cross-modal human understanding. To mitigate the computational complexity associated with integrating all features into a single model, we employ a multimodal and multi-teacher architecture.